

Weightwise (almost) perfectly balanced functions: t -concatenation and the general Maiorana-McFarland class

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Abstract

The study of cryptographic criteria for Boolean functions with restricted domains has been an important topic over the last 20 years. A revived interest has sparked after the work of Carlet, Méaux and Rotella in 2017, where the authors studied cryptographic properties of restricted-domain functions and introduced the concept of weightwise perfectly balanced functions as part of the analysis of the FLIP stream cipher. Weightwise (almost) perfectly balanced functions are defined as Boolean functions that are (almost) balanced on each of the sets of vectors of the same Hamming weight. Several approaches have been considered to build new families of such functions. In this article, we present some new constructions of weightwise (almost) perfectly balanced functions via two approaches, the first class is constructed using the t -concatenation of Boolean functions, whereas the second one draws certain functions from the so-called general Maiorana-McFarland class. A generic analysis of these two classes is given, as well as explicit examples in both classes. Namely, we provide instances of functions in both classes attaining high overall nonlinearities, as well as slice nonlinearities. Notably, we present examples in 16 variables that attain some of the best overall nonlinearities, and more importantly, the highest slice nonlinearities among all of the constructions presented in the literature.

Keywords: Boolean Functions, Concatenation, Generalized Maiorana-McFarland, Balanced Functions, Krawtchouk Polynomials, Weightwise Perfectly Balanced.

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1 Introduction

Cryptographic criteria of Boolean functions in restricted domains have been addressed in different works (see [2, 7–9, 13, 18]). In [3], the authors presented the first comprehensive study in this context, where they highlighted the importance of such functions in the filtering model of stream ciphers for which a suitable filtering functions is selected. In particular, a function with a restricted output is used for the FLIP cipher. An important conclusion drawn from

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the analysis in [3] is that the robustness evaluation of restricted-domain functions is a fundamental security aspect. Functions that are balanced over all subsets of $\mathbb{F}_2^n$ containing vectors with constant Hamming weight k , called *slices* and denoted by $E_{k,n}$, are said to be *weightwise perfectly balanced* (WPB). These functions exist only when n is a power of two. In general, when n is not a power of two, a function that is almost balanced on each subset of constant Hamming weight is said to be *weightwise almost perfectly balanced* (WAPB), where, in this context, the word *almost* means that the number of preimages of 0 and of 1 do not differ or they differ by at most 1, depending on the size of $E_{k,n}$. The definitions of weightwise perfectly balanced and weightwise almost perfectly balanced Boolean functions, as well as the first explicit constructions, were introduced in [3]. Soon after their introduction, several studies have focused on the construction of W(A)PB functions (see [4, 6, 12, 21, 27, 29, 31, 32]) and their cryptographic properties, where techniques such as concatenation [10], the direct sum [32], modification of support of affine functions, WAPBs and highly nonlinear functions [5, 16], or using the action of cyclic permutation groups [6] have been employed. Other algebraic methods exploiting different representations of Boolean functions have proved to be useful, e.g. the use of the trace form of rotation symmetric functions [15] or the multivariate representation of specific Boolean functions [30]. Algorithmic approaches have been presented recently in [19, 20, 28]. It is a well-established fact that the properties of balancedness and nonlinearity play an important role in the realm of cryptography, consequently, one of the most studied properties accompanying weightwise perfectly balanced functions is the *k-weightwise nonlinearity*, which is defined to be the distance between the restriction of a function to $E_{k,n}$ and the restriction of an affine function to the slice $E_{k,n}$. Many constructions presented in the literature yield functions that attain high *k-weightwise nonlinearities*, out of which the best overall nonlinearities can be found in [21]. Upper bounds on the weightwise nonlinearities of WAPB functions are known [3, 10, 24], however, the exact nonlinearities, and even lower bounds, are known only for a few families [10]. In [10], the authors proposed secondary constructions of WAPB functions whose weightwise nonlinearities can be lower bounded. The approach taken in [10] is based on Siegenthaler's construction (i.e. concatenation of Boolean functions) and the addition of symmetric functions whereby they built 8-variable and 16-variable WPB functions seeded with functions drawn from earlier families with known slice nonlinearities. A crucial notion for this construction is that of special weightwise almost perfectly balanced functions (SWAPB)—WAPB functions with fixed support size on slices of odd cardinality.

The aim of this paper is to enrich the subject of Boolean functions over restricted domains by presenting two general constructions of W(A)PB functions attaining high nonlinearities. Our first construction, a secondary construction, can be seen as a refinement of the ideas in [10]. Namely, for the concatenation f of WPB functions, we characterize when f is WAPB by deriving a natural combinatorial equivalent formulation. Then, we analyze the general case of concatenation of WAPB functions and derive sufficient conditions to obtain WAPB functions. These constructions can yield W(A)PB functions with rather high nonlinearities as illustrated by several instances. Thereafter, we introduce a method based on the general Maiorana-McFarland class to produce WAPB functions and explicitly derive their slice nonlinearities. To revise the scope of this method, we study the feasibility of obtaining all possible values for which one can obtain such functions for small spaces, i.e., $n \leq 16$. Notably, we exhibit functions in the GMM class that attain the best overall nonlinearity presented

in the literature so far, as well as very high slice nonlinearities, oftentimes, surpassing the values of many previous constructions. As a byproduct of our results and elaborating on the concept of special WAPB functions, we present a suitable notion of equivalence of functions, *slice equivalence*, which takes into account the effect of linear functions that either fix slices or swap them symmetrically. Intuitively, slice equivalence exploits the local nature of the WAPB property. In light of this equivalence, we propose a simple construction method to get new WAPB functions that is based on restriction functions. Finally, we formalize another approach of constructing WAPB functions, based on mirroring slices that have the best weightwise nonlinearities.

This article is structured as follows. In Section 2, we provide the necessary background and definitions that will be used throughout the article. The analysis and construction of weightwise almost perfectly balanced functions from the concatenation of weightwise perfectly balanced Boolean functions is given in Section 3, where we derive a combinatorial characterization of functions that yield WAPB functions. Afterwards, in Section 4, we study the general case of concatenation of weightwise almost perfectly balanced functions whereby we obtain sufficient conditions on the parameters of the concatenands that yield WAPB functions. A generic approach using the general Maiorana-McFarland class is introduced in Section 5. For this class of functions, we give a description of the possible parameters for which W(A)PB functions can be obtained. Additionally, the weightwise nonlinearities of these functions are computed and we present instances attaining the best overall nonlinearities as well as exceptionally high slice nonlinearities. In Section 6, we propose a new notion of equivalence based on slice linear functions acting on the input of a function together with the addition of symmetric functions. To illustrate the utility of this notion, a simple construction method is introduced, which yields new WAPB functions. Finally, some concluding remarks are given in Section 7.

2 Preliminaries

An n -variable Boolean function f is a function from $\mathbb{F}_2^n$ to $\mathbb{F}_2$. Let $\mathcal{B}_n$ denote the set of Boolean functions in n variables. For $x = (x_1, \dots, x_n)$, a Boolean function $f(x) \in \mathcal{B}_n$ is generally represented by its algebraic normal form (ANF):

$$f(x_1, \dots, x_n) = \sum_{b \in \mathbb{F}_2^n} \lambda_b \left(\prod_{i=1}^n x_i^{b_i} \right), \quad (1)$$

where $\lambda_b \in \mathbb{F}_2$, $b = (b_1, \dots, b_n) \in \mathbb{F}_2^n$. The algebraic degree of $f(x)$, denoted by $\deg(f)$, is the maximal value of $w_H(b)$ such that $\lambda_b \neq 0$, where $w_H(b)$ denotes the Hamming weight of b . A function $f \in \mathcal{B}_n$ is called an affine if $\deg(f) = 1$. An affine function with constant term equal to zero is called a linear function. Any linear function on $\mathbb{F}_2^n$ is denoted by:

$$\omega \cdot x = \omega_1 x_1 + \dots + \omega_n x_n,$$

where $\omega = (\omega_1, \dots, \omega_n)$, $x = (x_1, \dots, x_n) \in \mathbb{F}_2^n$. The Walsh transform of $f \in \mathcal{B}_n$ at the point ω is denoted by $W_f(\omega)$ and calculated as

$$W_f(\omega) = \sum_{x \in \mathbb{F}_2^n} (-1)^{f(x) + \omega \cdot x}. \quad (2)$$

The Walsh spectra of $f \in \mathcal{B}_n$ is defined as the multiset $\{W_f(\omega) \mid \omega \in \mathbb{F}_2^n\}$, where $W_f(\omega)$ is ordered lexicographically according to ω . The Walsh support of $f \in \mathcal{B}_n$ is defined as

$$S_f = \{\omega \in \mathbb{F}_2^n \mid W_f(\omega) \neq 0\}. \quad (3)$$

Let $\text{supp}(f) = \{x \in \mathbb{F}_2^n \mid f(x) = 1\}$ denote the support of f . A function $f \in \mathcal{B}_n$ is said to be balanced if its output column in the truth table contains equal number of 0s and 1s, i.e., $|\text{supp}(f)| = 2^{n-1}$, or equivalently, $W_f(0_n) = 0$. The nonlinearity of the Boolean function $f \in \mathcal{B}_n$ is characterized as

$$NL(f) = 2^{n-1} - \frac{1}{2} \max_{\omega \in \mathbb{F}_2^n} |W_f(\omega)|. \quad (4)$$

Denote $E_{k,n} = \{x \in \mathbb{F}_2^n : w_H(x) = k\}$ and $\text{supp}_k(f) = \{x \in E_{k,n} : f(x) = 1\} = \text{supp}(f) \cap E_{k,n}$. The Hamming distance of two functions $f, g \in \mathcal{B}_n$ on $E_{k,n}$ denoted as $d_{k,n}(f, g) = |\{x \in E_{k,n} : f(x) \neq g(x)\}|$. We reserve the notation $\chi_{k,n} : \mathbb{F}_2^n \rightarrow \mathbb{F}_2$ for the indicator (or characteristic) function of a given slice $E_{k,n}$, i.e., $\chi_{k,n}(x) = 1$ if and only if $w_H(x) = k$. Other cryptographic criteria like balancedness, nonlinearity and algebraic immunity of a function f defined over $\mathbb{F}_2^n$ can also be defined by restricting f to the set $E_{k,n}$. For two integers m, n with $m \leq n$, we define $[n] = \{1, \dots, n\}$ and $[m, n] = \{m, m+1, \dots, n\}$. Given a function $f \in \mathcal{B}_n$ and $k \in [n-1]$, we define

$$NL_k(f) = \frac{|E_{k,n}|}{2} - \frac{1}{2} \max_{\omega \in \mathbb{F}_2^n} |W_{f,k}(\omega)|,$$

where $W_{f,k}$ represents the Walsh transform of f restricted to $E_{k,n}$ at $\omega \in \mathbb{F}_2^n$.

Definition 2.1 Let $n = 2^m$, $m \geq 1$, and let $f \in \mathcal{B}_n$ such that $f(0_n) = f(1_n) + 1$. We call f perfectly weightwise balanced (WPB) if it is balanced on $E_{k,n}$, for each $k = 1, \dots, n-1$. In other words, f is WPB if

$$|\text{supp}_k(f)| = \frac{|E_{k,n}|}{2} = \frac{\binom{n}{k}}{2}, \quad \forall k \in \{1, \dots, n-1\}.$$

When discussing weightwise nonlinearity and balancedness, we can always assume without loss of generality that $f(0_n) = 0$ and $f(1_n) = 1$ since all relevant properties are preserved after swapping.

Definition 2.2 A function $f \in \mathcal{B}_n$ is called almost perfectly weightwise balanced (WAPB) if

$$|\text{supp}_k(f)| = \begin{cases} \frac{|E_{k,n}|}{2}, & \text{if } |E_{k,n}| \text{ is even,} \\ \frac{|E_{k,n}| \pm 1}{2}, & \text{if } |E_{k,n}| \text{ is odd.} \end{cases}$$

In this case, the boundary conditions on $f(0_n)$ and $f(1_n)$ are dropped. It is evident that the class of WPB functions is included in the class of WAPB functions.

Definition 2.3 *Let $t = 2^l$. Consider $f_0, f_1, \dots, f_{t-1} \in \mathcal{B}_n$, we define the t -concatenation of these functions, denoted by $f = f_0 || f_1 || \dots || f_{t-1}$, as the function $f \in \mathcal{B}_{n+l}$ defined recursively as follows:*

- (2-concatenation) For $l = 1$, $f = f_0 || f_1$ is the $(n+1)$ -variable function whose canonical subfunctions are f_0 and f_1 , i.e., $f(x, 0) = f_0(x)$ and $f(x, 1) = f_1(x)$ for every $x \in \mathbb{F}_2^n$.
- For $l > 1$, $f = f_0 || \dots || f_{t-1}$ is the 2-concatenation of $f_0 || \dots || f_{\frac{t}{2}-1}$ and $f_{\frac{t}{2}} || \dots || f_{t-1}$.

The 2-concatenation (or, simply, concatenation) of Boolean functions was introduced by Siegenthaler [25] as a secondary constructions of resilient functions. This technique has been widely used in other constructions of cryptographic significant functions. The special case for $t = 4$ has also been extensively studied in the context of bent decompositions (see [1]).

We recall two important identities from number theory, which will be useful throughout the following sections.

Proposition 2.1 (Pascal's identity) *Let $n, k \in \mathbb{N}$. Then the following holds*

$$\binom{n}{k} = \binom{n}{k-1} + \binom{n-1}{k-1}.$$

Proposition 2.2 (Vandermonde's identity) *Let $m, n, k \in \mathbb{N}$. Then the following holds*

$$\binom{m+n}{k} = \sum_{i=1}^k \binom{m}{i} \binom{n}{k-i}.$$

Every integer $0 \leq i \leq 2^l - 1$ can be written as $i = \sum_{j=0}^{l-1} i_j 2^j$, where $i_j \in \{0, 1\}$. The sequence $(i_0, \dots, i_{l-1})$ is called the *binary expansion* of the integer i , denoted by $\text{bin}(i)$. Note that the power l will not be explicitly displayed since it will always be clear from context. Given two vectors $u, v \in \mathbb{F}_2^n$, we say that u covers v , denoted by $v \preceq u$, provided that $\text{supp}(v) \subseteq \text{supp}(u)$. When discussing integers, the covering relation is understood as the covering relation of their binary expansions. Lucas' Theorem says that $k \preceq n$ iff the binomial coefficient $\binom{n}{k}$ is odd, we state it in the following proposition to further refer to it.

Proposition 2.3 (Lucas' Theorem) *Let $n, k \in \mathbb{N}$. Then $\binom{n}{k}$ is odd iff $k \preceq n$.*

For a given vector $v \in \mathbb{F}_2^n$ and a set of indices $I \subseteq [n]$, we will use the notation v^I to refer to the vector v in $\mathbb{F}_2^{n-|I|}$ punctured at entries indexed by I . Given two vectors a and b , we represent their concatenation by $a \frown b$. Numbers will sometimes be considered as vectors of length one.

Let m be a positive integer and let x be a variable taking non-negative integer values. Recall that the Krawtchouk polynomials [14] of order m are defined by

$$P_i(x, m) = \sum_{j=0}^i (-1)^j \binom{x}{j} \binom{m-x}{i-j} \quad (5)$$

where $0 \leq i \leq m$. Whenever there is no room for ambiguity, we write $P_i(x) := P_i(x, m)$, for simplicity. From the properties of Krawtchouk polynomials, we know

$$P_i(j) = (-1)^j P_{m-i}(j), \quad 0 \leq j \leq m. \quad (6)$$

The next result follows easily from the well-known properties of Krawtchouk polynomials (see, for instance, [17]).

Lemma 2.1 *Let $u \in \mathbb{F}_2^m$ with Hamming weight $wt(u) = j$. Then*

$$\sum_{v \in E_{i,m}} (-1)^{u \cdot v} = P_i(j).$$

3 Concatenation of WPB functions

In this section, we present the first construction of W(A)PB functions using the concatenation of Boolean functions. We characterize precisely when one can obtain weightwise perfectly balanced functions in this setting. Note that concatenation of Boolean functions has been employed as a secondary construction to construct W(A)PB functions, see [10]. Our approach can then be viewed as an extension of these results.

From the definition of t -concatenation, one can obtain its algebraic normal form using a simple inductive argument.

Proposition 3.1 *(Algebraic normal form of t -concatenation) Let $n \in \mathbb{N}$ and $t = 2^l$, where $l \leq n$. The t -concatenation of functions $f_0, f_1, \dots, f_{t-1} \in \mathcal{B}_n$ has the following algebraic normal form*

$$f(x, x_{n+1}, \dots, x_{n+l}) = \left(\bigparallel_{i=0}^{t-1} f_i \right) (x, x_{n+1}, \dots, x_{n+l}) = \sum_{i=0}^{t-1} f_{t-1-i}(x) \prod_{j=0}^{l-1} (x_{n+j+1} + i_j),$$

where $x \in \mathbb{F}_2^n$ and $(i_0, \dots, i_{l-1})$ is the binary expansion of i .

Although a similar approach can be carried out for $l > n$, the previous characterization keeps the length of the binary expansion constant, which will simplify our discussion.

Before diving into the general case, it is instructive to analyze the case $t = 4$. For this, let $n = 2^m$, $m \geq 2$, and $f_0, f_1, f_2, f_3 \in \mathcal{B}_n$ be WPB functions. Consider their 4-concatenation f , i.e.,

$$f(x, x_{n+1}, x_{n+2}) = (f_0 || f_1 || f_2 || f_3)(x, x_{n+1}, x_{n+2}),$$

whose ANF is

$$(x_{n+1} + 1)(x_{n+2} + 1)f_0(x) + x_{n+1}(x_{n+2} + 1)f_1(x) + (x_{n+1} + 1)x_{n+2}f_2(x) + x_{n+1}x_{n+2}f_3(x).$$

We claim that f is WAPB if $f_1(0_n) = f_2(0_n) + 1$, equivalently, $f_1(1_n) = f_2(1_n) + 1$ as f_1 and f_2 are WPB. Indeed, since it holds

$$|\text{supp}_1(f)| = |\text{supp}_1(f_0)| + |\text{supp}_0(f_1)| + |\text{supp}_0(f_2)| = \frac{n}{2} + 1 = \frac{1}{2} \binom{n+2}{1}$$

and

$$\begin{aligned}
|\text{supp}_2(f)| &= |\text{supp}_2(f_0)| + |\text{supp}_1(f_1)| + |\text{supp}_1(f_2)| + |\text{supp}_0(f_3)| \\
&= \frac{1}{2} \binom{n}{2} + \binom{n}{1} + |\text{supp}_0(f_3)| \\
&= \frac{1}{2} \binom{n+2}{2} - \frac{1}{2} + |\text{supp}_0(f_3)| \\
&= \frac{1}{2} \left(\binom{n+2}{2} \pm 1 \right),
\end{aligned}$$

where the symbol is $+$ if $f_3(0_n) = 1$ and $-$, otherwise. Due to the symmetry of the binomial coefficients, we get similar values for $|\text{supp}_{n+1}(f)|$ and $|\text{supp}_n(f)|$, respectively. Moreover, for $3 \leq k \leq n-1$, it holds

$$\begin{aligned}
|\text{supp}_k(f)| &= |\text{supp}_k(f_0)| + |\text{supp}_{k-1}(f_1)| + |\text{supp}_{k-1}(f_2)| + |\text{supp}_{k-2}(f_3)| \\
&= \frac{1}{2} \binom{n}{k} + \binom{n}{k-1} + \frac{1}{2} \binom{n}{k-2} \\
&= \frac{1}{2} \binom{n+1}{k} + \frac{1}{2} \binom{n+1}{k-1} \\
&= \frac{1}{2} \binom{n+2}{k}.
\end{aligned}$$

We conclude the 4-concatenation f is WAPB.

Now, let us turn to the general case. Let $x' = (x_{n+1}, \dots, x_{n+l})$. We say f_i is *activated* by x' if $f(x, x') = f_i(x)$. By $\mathcal{F}^{(j)}$ we denote the set of functions which are activated by some x' for which it holds $w_H(x') = j$, $j = 0, \dots, l$. It is easy to see that in $\mathcal{F}^{(j)}$ will be all functions f_i for which it holds that there are j nonzero bits in $\text{bin}(i)$. Note that $\{\mathcal{F}^{(j)}\}_{j=0, \dots, l}$ is a partition of $\{f_0, \dots, f_{t-1}\}$. It holds $|\mathcal{F}^{(j)}| = \binom{l}{j}$. By $f^{(j)}$ we denote a representative of each set $\mathcal{F}^{(j)}$ that can be defined as $f^{(j)} = f_{\min_{f_i \in \mathcal{F}^{(j)}} i}$. From the definition of t -concatenation, we can write the k -support of f , for each $k \in [n]$, as follows

$$|\text{supp}_k(f)| = \sum_{j=0}^l \sum_{g \in \mathcal{F}^{(j)}} |\text{supp}_{k-j}(g)|. \quad (7)$$

Theorem 3.1 *Let $n > 2$ be a power of two and $f_0, \dots, f_{t-1} \in \mathcal{B}_n$ be WPB functions, $t = 2^l$, $l < n$, and let $f \in \mathcal{B}_{n+l}$ be the t -concatenation of $f_0, \dots, f_{t-1}$. The following statements are equivalent:*

- (1) *For any $j \in [l-1]$, the number of functions from $\mathcal{F}^{(j)}$ that map 0_n to 0 is $\frac{|\mathcal{F}^{(j)}|}{2}$ whenever $\binom{l}{j}$ is even, and $\frac{|\mathcal{F}^{(j)}| \pm 1}{2}$ otherwise;*
- (2) *The function f is WAPB.*

Proof. Suppose that condition (1) in the statement holds and let $k \in [n + l - 1]$.

For $k \in [l]$, from (7),

$$|\text{supp}_k(f)| = \sum_{j=0}^{k-1} \binom{l}{j} |\text{supp}_{k-j}(f^{(j)})| + \sum_{g \in \mathcal{F}^{(k)}} |\text{supp}_0(g)|.$$

Since all concatenands are WPB, the above expression is equal to

$$\begin{aligned} |\text{supp}_k(f)| &= \frac{1}{2} \sum_{j=0}^{k-1} \binom{l}{j} \binom{n}{k-j} + \sum_{g \in \mathcal{F}^{(k)}} |\text{supp}_0(g)| \\ &= \frac{1}{2} \left(\sum_{j=0}^k \binom{l}{j} \binom{n}{k-j} - \binom{l}{k} \binom{n}{0} \right) + \sum_{g \in \mathcal{F}^{(k)}} |\text{supp}_0(g)| \\ &= \frac{1}{2} \left(\binom{n+l}{k} - \binom{l}{k} \right) + \sum_{g \in \mathcal{F}^{(k)}} |\text{supp}_0(g)|. \end{aligned} \quad (8)$$

So $\binom{n+l}{k}$ and $\binom{l}{k}$ have the same parity. If $\binom{l}{k}$ is even, then $\sum_{g \in \mathcal{F}^{(k)}} |\text{supp}_0(g)| = \frac{|\mathcal{F}^{(k)}|}{2} = \frac{1}{2} \binom{l}{k}$, and (8) becomes

$$|\text{supp}_k(f)| = \frac{1}{2} \left(\binom{n+l}{k} - \binom{l}{k} \right) + \frac{1}{2} \binom{l}{k} = \frac{1}{2} \binom{n+l}{k}.$$

If $\binom{l}{k}$ is odd, then $\sum_{g \in \mathcal{F}^{(k)}} |\text{supp}_0(g)| = \frac{|\mathcal{F}^{(k)}| \pm 1}{2} = \frac{1}{2} \left(\binom{l}{k} \pm 1 \right)$, and (8) becomes

$$|\text{supp}_k(f)| = \frac{1}{2} \left(\binom{n+l}{k} - \binom{l}{k} \right) + \frac{1}{2} \left(\binom{l}{k} \pm 1 \right) = \frac{1}{2} \left(\binom{n+l}{k} \pm 1 \right).$$

For $k \in [l + 1, n - 1]$, $\binom{n+l}{k}$ is even since each summand $\binom{n}{k-j}$ is even, for all $j = 0, \dots, k$, by Proposition 2.2. From (7), we get

$$|\text{supp}_k(f)| = \sum_{j=0}^l \binom{l}{j} |\text{supp}_{k-j}(f^{(j)})| = \frac{1}{2} \sum_{j=0}^l \binom{l}{j} \binom{n}{k-j} = \frac{1}{2} \binom{n+l}{k}. \quad (9)$$

For $k \in [n, n + l - 1]$, set $k = n + r$, where $r \in [0, l - 1]$. After a simple manipulation of binomial coefficients, we get

$$|\text{supp}_k(f)| = |\text{supp}_{n+r}(f)| = \sum_{g \in \mathcal{F}^{(r)}} |\text{supp}_n(g)| + \frac{1}{2} \left(\binom{n+l}{k} - \binom{l}{r} \right). \quad (10)$$

Note that condition (1) implies that the number of functions in $\mathcal{F}^{(j)}$, for each $j \in [l - 1]$, that map 1_n to 1 is $\frac{1}{2} \binom{l}{r}$ provided that $\binom{l}{r}$ is even and $\frac{1}{2} \left(\binom{l}{r} \pm 1 \right)$ when $\binom{l}{r}$ is odd because all concatenands are WPB. Hence, if $\binom{l}{r}$ is even, then $\sum_{g \in \mathcal{F}^{(r)}} |\text{supp}_n(g)| = \frac{|\mathcal{F}^{(r)}|}{2} = \frac{1}{2} \binom{l}{r}$,

and then $|\text{supp}_k(f)| = \frac{1}{2} \binom{n+l}{k}$ by (10). If $\binom{l}{r}$ is odd, then $\sum_{g \in \mathcal{F}(r)} |\text{supp}_n(g)| = \frac{|\mathcal{F}(r)| \pm 1}{2} = \frac{1}{2} \left(\binom{l}{r} \pm 1 \right)$, and then $|\text{supp}_k(f)| = \frac{1}{2} \left(\binom{n+l}{k} \pm 1 \right)$ by (10). We conclude f is WAPB.

Conversely, suppose that f is WAPB. Let $j \in [l]$. From (8), we know that $\binom{n+l}{j}$ and $\binom{l}{j}$ have the same parity and

$$\sum_{g \in \mathcal{F}(j)} |\text{supp}_0(g)| = |\text{supp}_j(f)| - \frac{1}{2} \left(\binom{n+l}{j} - \binom{l}{j} \right).$$

If $\binom{l}{j}$ is even (thus $\binom{n+l}{j}$ is even), then $|\text{supp}_j(f)| = \frac{1}{2} \binom{n+l}{j}$. Hence, $\sum_{g \in \mathcal{F}(j)} |\text{supp}_0(g)| = \frac{1}{2} \binom{l}{j}$. Similarly, if $\binom{l}{j}$ is odd, then $\sum_{g \in \mathcal{F}(j)} |\text{supp}_0(g)| = \frac{1}{2} \left(\binom{l}{j} \pm 1 \right)$ by (8) and the fact that f is WAPB. This implies elements in $\mathcal{F}^{(j)}$ that map 0_n to 0 satisfy condition (1). $\square$

Theorem 3.2 *Let $n > 2$ be a power of two and $f_0, \dots, f_{t-1} \in \mathcal{B}_n$ be WPB functions, where $t = 2^n$. Let $f \in \mathcal{B}_{2n}$ be the t -concatenation of $f_0, \dots, f_{t-1}$. The following statements are equivalent:*

(1) *For any $j \in [n-1]$, the number of functions from $\mathcal{F}^{(j)}$ that map 0_n to 0 is $\frac{|\mathcal{F}^{(j)}|}{2}$ and $f_0(0_n) = f_{t-1}(1_n) + 1$;*

(2) *f is WPB.*

Proof. Since n is a power of 2, so is $2n$ and every binomial coefficient $\binom{2n}{k}$ is even. Suppose that (1) holds. Using similar arguments as in the proof of Theorem 3.1, one can show that for $k \in [n-1]$ and $k \in [n+1, 2n-1]$, we have $|\text{supp}_k(f)| = \frac{1}{2} \binom{2n}{k}$. The only value we need to examine is $|\text{supp}_n(f)|$. Compute

$$\begin{aligned} |\text{supp}_n(f)| &= \sum_{j=0}^n \sum_{g \in \mathcal{F}^{(j)}} |\text{supp}_{n-j}(g)| \\ &= |\text{supp}_n(f_0)| + \sum_{j=1}^{n-1} \binom{n}{j} |\text{supp}_{n-j}(g)| + |\text{supp}_0(f_{t-1})| \\ &= |\text{supp}_n(f_0)| + \sum_{j=1}^{n-1} \binom{n}{j} \binom{n}{n-j} + |\text{supp}_0(f_{t-1})| \\ &= |\text{supp}_n(f_0)| + \frac{1}{2} \left(\binom{2n}{n} - 2 \right) + |\text{supp}_0(f_{t-1})|. \end{aligned}$$

Since $f_0(0_n) = f_{t-1}(1_n) + 1$,

$$|\text{supp}_n(f)| = 1 + \frac{1}{2} \left(\binom{2n}{n} - 2 \right) = \frac{1}{2} \binom{2n}{n}.$$

Therefore, f is WPB. Suppose that f is WPB. By definition of weightwise perfect balancedness,

$$f_0(0_n) = f(0_n, 0_n) = f(1_n, 1_n) + 1 = f_{2^n-1}(1_n) + 1.$$

Let $j \in [n - 1]$. From (7), we deduce that

$$\sum_{g \in \mathcal{F}^{(j)}} |\text{supp}_0(g)| = |\text{supp}_j(f)| - \frac{1}{2} \left(\binom{2n}{j} - \binom{n}{j} \right). \quad (11)$$

Since $\binom{2n}{j}$ is even, $|\text{supp}_j(f)| = \frac{1}{2} \binom{2n}{j}$. Hence, $\sum_{g \in \mathcal{F}^{(j)}} |\text{supp}_0(g)| = \frac{1}{2} \binom{n}{j}$. This yields that the elements in $\mathcal{F}^{(j)}$ that map 0_n to zero satisfy condition (1). $\square$

Example 3.1 Let $h_2 \in \mathcal{B}_4$ be a WPB function as defined in [30]. Then the 16-concatenation $f = h_2 + 1 || h_2 + 1 || h_2 || h_2 || h_2 || h_2 + 1 || h_2 || h_2 + 1 || h_2 || h_2 + 1 || h_2 || h_2 + 1 || h_2 + 1 || h_2 + 1$ is an 8-variable WPB function with algebraic normal form

$$\begin{aligned} f(x) = & x_0 x_1 x_2 + x_0 x_2 x_3 + x_0 x_2 + x_0 x_1 + x_1 x_2 + x_1 + x_0 + 1 \\ & + x_4 x_5 x_6 x_7 + x_5 x_6 x_7 + x_4 x_5 + x_4 x_5 x_7 + x_4 x_6 + x_4 x_7 + x_5 + x_6, \end{aligned}$$

and $NL(f) = 104$. Its slice nonlinearities are $NL_2(f) = 6$, $NL_3(f) = 14$, $NL_4(f) = 17$, $NL_5(f) = 15$ and $NL_6(f) = 6$.

Notice that the function f in Example 3.1 could be seen as a direct sum of two WPB functions—this technique has already been used for obtaining W(A)PB functions [32]. The next example exhibits a WPB from t -concatenation, which does not overlap with this class.

Example 3.2 Let $f_2, h_2 \in \mathcal{B}_4$ be WPB functions defined in [30]. The 16-concatenation

$f = f_2 || h_2 + 1 || h_2 || h_2 || h_2 || h_2 + 1 || h_2 || h_2 + 1 || h_2 || h_2 + 1 || h_2 || h_2 + 1 || h_2 + 1 || f_2 + 1$ is an 8-variable WPB function, whose algebraic normal form is

$$\begin{aligned} f(x) = & x_0 x_1 x_2 + x_0 x_1 x_4 x_5 x_6 + x_0 x_1 x_4 x_5 x_7 + x_0 x_1 x_4 x_5 + x_0 x_1 x_4 x_6 x_7 + x_0 x_1 x_4 x_6 \\ & + x_0 x_1 x_4 x_7 + x_0 x_1 x_4 + x_0 x_1 x_5 x_6 x_7 + x_0 x_1 x_5 x_6 + x_0 x_1 x_5 x_7 + x_0 x_1 x_5 + x_0 x_1 x_6 x_7 \\ & + x_0 x_1 x_6 + x_0 x_1 x_7 + x_0 x_2 x_3 + x_0 x_2 + x_0 + x_1 x_2 x_4 x_5 x_6 + x_1 x_2 x_4 x_5 x_7 + x_1 x_2 x_4 x_5 \\ & + x_1 x_2 x_4 x_6 x_7 + x_1 x_2 x_4 x_6 + x_1 x_2 x_4 x_7 + x_1 x_2 x_4 + x_1 x_2 x_5 x_6 x_7 + x_1 x_2 x_5 x_6 + x_4 x_6 x_7 \\ & + x_1 x_2 x_5 x_7 + x_1 x_2 x_5 + x_1 x_2 x_6 x_7 + x_1 x_2 x_6 + x_1 x_2 x_7 + x_1 + x_4 x_5 x_6 x_7 + x_4 x_5 x_6 \\ & + x_4 + x_5 x_6 + x_5 x_7 + x_6 x_7 + x_7 \end{aligned}$$

and $NL(f) = 104$. Moreover, $NL_2(f) = 4$, $NL_3(f) = 14$, $NL_4(f) = 17$, $NL_5(f) = 15$, and $NL_6(f) = 4$.

It is important to note that concatenation, that is, Siegenthaler's construction [25], has already been used for constructing W(A)PB mappings, in particular, in the context of so-called *special WAPB* functions (SWAPB), firstly introduced in [10].

Definition 3.1 The function $f \in \mathcal{B}_n$ is called *special WAPB (SWAPB)* if

$$|\text{supp}_k(f)| = \begin{cases} \frac{|E_{k,n}|}{2}, & \text{if } |E_{k,n}| \text{ is even,} \\ \frac{|E_{k,n}|+1}{2}, & \text{if } |E_{k,n}| \text{ is odd and } k < \frac{n}{2}. \\ \frac{|E_{k,n}|-1}{2}, & \text{if } |E_{k,n}| \text{ is odd and } k > \frac{n}{2}. \end{cases}$$

In [10], it was shown that any WAPB function can be used together with a SWAPB function to yield a SWAPB function from 2-concatenation. To demonstrate the difference between this construction and the generation of WAPB functions via t -concatenation of W(A)PB functions, we provide examples to show that 2-concatenation of non-special WAPB functions can result in WPB functions (which are by definition special) that are not covered by the results of [10]. Moreover, 2-concatenation of non-special WAPB functions can also yield a non-special WAPB function that is inequivalent to special functions from Theorem 1 in [10].

Example 3.3 Let $g_2 \in \mathcal{B}_4$ be the WPB function defined in [22], and let

$$f_0 = g_2 + 1||g_2||g_2||g_2||g_2 + 1||g_2||g_2 + 1||g_2$$

and

$$f_1 = g_2 + 1||g_2||g_2 + 1||g_2||g_2 + 1||g_2 + 1||g_2 + 1||g_2.$$

It holds that functions $f_0, f_1 \in \mathcal{B}_7$ are WAPB but are not SWAPB. Their concatenation $f = f_0||f_1$ is a WPB 8-variable Boolean function with nonlinearity 96. By adding suitable slice indicators, the functions f_0 and f_1 can be turned into special functions f'_0 and f'_1 . Applying the construction given in Theorem 1 of [10], we obtain a WPB function f' (thus special) from the concatenation of f_0 and g , where

$$g = f'_1 + \sum_{i \in [1,4]} \chi_{i-1,7} + \chi_{7-i,7}.$$

The overall nonlinearity of f' is 84, which implies that f and f' are different functions. Moreover, they are inequivalent functions since $NL_6(f) = 4$ and $NL_6(f') = 5$.

Example 3.4 Let $g_3 \in \mathcal{B}_8$ be defined as in [22], and let

$$f_0 = g_3 + 1||g_3||g_3||g_3||g_3 + 1||g_3||g_3 + 1||g_3$$

and

$$f_1 = g_3 + 1||g_3||g_3 + 1||g_3||g_3 + 1||g_3 + 1||g_3 + 1||g_3.$$

It holds that functions $f_0, f_1 \in \mathcal{B}_{11}$ are WAPB but are not SWAPB. Their concatenation $f = f_0||f_1$ is a WAPB 12-variable Boolean function with nonlinearity 1915, which is not SWAPB. The function f can be turned into an SWAPB with nonlinearity 1872 by adding suitable slice indicators. Similarly, the functions f_0 and f_1 can be made special functions, say, f'_0 and f'_1 . The corresponding SWAPB function $f' = f_0||g$, where

$$g = f'_1 + \sum_{i \in [1,3]} \chi_{i-1,11} + \chi_{7-i,11}$$

has nonlinearity 1852. Moreover, $NL_3(f) = 57, NL_8(f) = 192, NL_{10}(f) = 5$, and $NL_3(f') = 58, NL_8(f') = 176, NL_{10}(f') = 10$, respectively. Therefore, f is inequivalent to f' .

3.1 Walsh values of the t -concatenation

Consider two functions $f_0, f_1 \in \mathcal{B}_n$. By definition,

$$f(x, x_{n+1}) = (x_{n+1} + 1)f_0(x) + x_{n+1}f_1(x),$$

which implies $f(x, 0) = f_0(x)$ and $f(x, 1) = f_1(x)$. The Walsh transform of $f \in \mathcal{B}_n$ at point (u_0, u_1) is given by

$$\begin{aligned} W_f(u_0, u_1) &= \sum_{x \in \mathbb{F}_2^n} (-1)^{f(x, x_{n+1}) \oplus (x, x_{n+1}) \cdot (u_0, u_1)} \\ &= \sum_{x \in \mathbb{F}_2^n} (-1)^{f_0(x) \oplus (x, 0) \cdot (u_0, u_1)} + \sum_{x \in \mathbb{F}_2^n} (-1)^{f_1(x) \oplus (x, 1) \cdot (u_0, u_1)} \\ &= \sum_{x \in \mathbb{F}_2^n} (-1)^{f_0(x) \oplus x \cdot u_0} + (-1)^{u_1} \sum_{x \in \mathbb{F}_2^n} (-1)^{f_1(x) \oplus x \cdot u_0} \\ &= W_{f_0}(u_0) + (-1)^{u_1} W_{f_1}(u_0). \end{aligned}$$

In general, the Walsh values of a t -concatenation of Boolean functions is given by a sum of Walsh values of the constituent functions as shown in the next proposition.

Proposition 3.2 *Let $f_0, \dots, f_{t-1} \in \mathcal{B}_n$ be given. Then the Walsh transform of the t -concatenation $f_0 || \dots || f_{t-1} \in \mathcal{B}_{n+l}$ at a point $(u_0, u_1, \dots, u_l)$ is*

$$\sum_{j=0}^{2^l-1} (-1)^{\text{bin}(j) \cdot (u_1, \dots, u_l)} W_{f_j}(u_0).$$

Proof. Let $f_0 || \dots || f_{t-1} \in \mathcal{B}_n$ be given. Since $t = 2^l$, we can write the t -concatenation $f \in \mathcal{B}_{n+l}$ as the 2-concatenation $g || g'$, where $g = f_0 || \dots || f_{t/2-1}$ and $g' = f_{t/2} || \dots || f_{t-1}$. Then,

$$W_f(u_0, u_1, \dots, u_l) = W_g(u_0, \dots, u_{l-1}) + (-1)^{u_l} W_{g'}(u_0, \dots, u_{l-1}).$$

By an inductive argument, we get that $W_f(u_0, u_1, \dots, u_l)$ is equal to

$$\sum_{j=0}^{2^{l-1}-1} (-1)^{\text{bin}(j) \cdot (u_1, \dots, u_{l-1})} W_{f_j}(u_0) + (-1)^{u_l} \sum_{j=2^{l-1}}^{2^l-1} (-1)^{\text{bin}(j) \cdot (u_1, \dots, u_{l-1})} W_{f_j}(u_0).$$

This expression can be turned into

$$\sum_{j=0}^{2^{l-1}-1} (-1)^{(\text{bin}(j) \cap 0) \cdot (u_1, \dots, u_{l-1}, u_l)} W_{f_j}(u_0) + \sum_{j=2^{l-1}}^{2^l-1} (-1)^{(\text{bin}(j) \cap 1) \cdot (u_1, \dots, u_{l-1}, u_l)} W_{f_j}(u_0).$$

Equivalently, $W_f(u_0, u_1, \dots, u_l) = \sum_{j=0}^{2^l-1} (-1)^{\text{bin}(j) \cdot (u_1, \dots, u_l)} W_{f_j}(u_0)$. $\square$

For instance, let us consider the 4-concatenation of functions $f_0, f_1, f_2, f_3 \in \mathcal{B}_n$. The Walsh transform of $f_0 || f_1 || f_2 || f_3 \in \mathcal{B}_{n+2}$ at a point (u_0, u_1, u_2) can be computed as

$$W_{f_0}(u_0) + (-1)^{u_1} W_{f_1}(u_0) + (-1)^{u_2} W_{f_2}(u_0) + (-1)^{u_1+u_2} W_{f_3}(u_0),$$

which is in accordance with Proposition 3.2.

4 The general case

Thus far, we have considered the problem of concatenating WPB functions to obtain a new W(A)PB function. In this section, we analyze the more general case when the constituent functions are W(A)PB.

Let us suppose that $f = f_0 || f_1 \in \mathcal{B}_{n+1}$ is a WAPB function, i.e., for each k ,

$$|\text{supp}_k(f)| = \frac{\binom{n+1}{k} + \epsilon_k}{2},$$

where $\epsilon_k \in \{-1, 0, 1\}$. Since $|\text{supp}_k(f)| = |\text{supp}_k(f_0)| + |\text{supp}_{k-1}(f_1)|$, these relations impose some restrictions on the functions f_0 and f_1 . For $n \in \mathbb{N}$, define the vector $\gamma^{(n)} = (\binom{n}{j} \pmod{2})_{0 \leq j \leq n} \in \mathbb{F}_2^{n+1}$, i.e., the vector $\gamma^{(n)}$ codifies the parity of the binomial coefficients $\binom{n}{j}$. By restating the definition of WAPB functions for the case of the concatenation of two functions, we obtain the following simple characterization, whose proof is omitted.

Proposition 4.1 *Let $n \in \mathbb{N}$. Let $\varepsilon = (\epsilon_0, \dots, \epsilon_{n+1}) \in \{-1, 0, 1\}^{n+2}$ be a vector such that $\text{supp}(\varepsilon) = \text{supp}(\gamma^{(n+1)})$. Let $(l_0, \dots, l_n), (m_0, \dots, m_n) \in \mathbb{N}^{n+1}$ be two integer sequences such that*

- i) For each $j \in [0, n]$, $0 \leq l_j, m_j \leq \binom{n}{j}$;*
- ii) $l_0 = m_n = 0$;*
- iii) For each $j \in [1, n]$, $l_j + m_{j-1} = \frac{\binom{n+1}{j} + \epsilon_j}{2}$.*

Let f_0, f_1 be two Boolean functions such that $\text{wt}(f_0 \upharpoonright_{E_{j,n}}) = l_j$ and $\text{wt}(f_1 \upharpoonright_{E_{j,n}}) = m_j$. Then the function $f = f_0 || f_1$ is WAPB. Conversely, if a function $f \in \mathcal{B}_{n+1}$ is WAPB and $f = f_0 || f_1$ for some $f_0, f_1 \in \mathcal{B}_n$, then there are sequences of integers $(\epsilon_0, \epsilon_1, \dots, \epsilon_{n+1}), (l_0, \dots, l_n)$, and $(m_0, \dots, m_n)$ satisfying i), ii) and iii).

From the previous proposition, it is now evident that the building blocks f_0 and f_1 need not be WAPB. Conversely, as in the WPB case, not every two WAPB functions can be concatenated to get a WAPB function. Therefore, the goal is now to derive some sufficient conditions that assure the resulting functions is weightwise almost perfectly balanced.

Definition 4.1 *Let $f \in \mathcal{B}_n$ and $k \in \mathbb{N}$ with $0 \leq k \leq n$. The function f is called k -almost perfectly weightwise balanced of type ε (k -WAPB $^{(\varepsilon)}$) if*

$$\varepsilon = 2|\text{supp}_k(f)| - \binom{n}{k} \in \{-1, 0, 1\}.$$

When $\varepsilon = 0$, we also use the term k -perfectly weightwise balanced (k -WPB). We will employ the convention that every Boolean function is 0-WPB and n -WPB.

The set of k -WAPB⁽¹⁾ (resp. k -WAPB⁽⁻¹⁾, k -WPB = k -WAPB⁽⁰⁾) functions in n -variables is denoted by $\mathcal{WAPB}_{k,n}^+$ (resp. $\mathcal{WAPB}_{k,n}^-$, $\mathcal{WPB}_{k,n} = \mathcal{WAPB}_{k,n}^0$). In this context, we will often make an abuse of notation by simply replacing $+$ with 1 , $-$ with -1 , and vice versa. Moreover, the subscript n is omitted whenever it is clear from the context.

Thus, the set of WAPB functions in n variables is

$$\mathcal{WAPB}_n = \left(\bigcap_{k \not\leq n} \mathcal{WPB}_{k,n} \right) \cap \left(\bigcap_{k \leq n} \mathcal{WAPB}_{k,n}^- \cup \mathcal{WAPB}_{k,n}^+ \right).$$

Proposition 4.2 *Let $n \in \mathbb{N}$ be even and $f_0, f_1 \in \mathcal{B}_n$. Define $f \in \mathcal{B}_{n+1}$ to be the concatenation of f_0 with f_1 , i.e. $f = f_0 || f_1$. The following hold for $k \in [2, n-1]$ and $(a, b, c) \in \{0, +, -\}^3$.*

1) *If k is even, then only for either $(a, b, c) = (0, 0, 0)$ or $(a, b, c) = (\pm, \pm, 0)$, it holds*

$$f \in \mathcal{WAPB}_{k,n+1}^a \text{ if } f_0 \in \mathcal{WAPB}_{k,n}^b \text{ and } f_1 \in \mathcal{WAPB}_{k-1,n}^c.$$

2) *If k is odd, then only for either $(a, b, c) = (0, 0, 0)$ or $(a, b, c) = (\pm, 0, \pm)$, it holds*

$$f \in \mathcal{WAPB}_{k,n+1}^a \text{ if } f_0 \in \mathcal{WAPB}_{k,n}^b \text{ and } f_1 \in \mathcal{WAPB}_{k-1,n}^c.$$

Proof. Suppose that $f_0 \in \mathcal{WAPB}_{k,n}^b$ and $f_1 \in \mathcal{WAPB}_{k-1,n}^c$. The set $\text{supp}_k(f)$ satisfies $|\text{supp}_k(f)| = \frac{\binom{n+1}{k} + b + c}{2}$ since

$$|\text{supp}_k(f)| = |\text{supp}_k(f_0)| + |\text{supp}_{k-1}(f_1)|$$

and

$$\frac{\binom{n}{k} + b}{2} + \frac{\binom{n}{k-1} + c}{2} = \frac{\binom{n+1}{k} + b + c}{2}$$

by Pascal's identity. Set $a := b + c$. Let us determine the triplets (a, b, c) for which $f \in \mathcal{WAPB}_{k,n+1}^a$. To do so, we will examine all feasible solutions according to the parity of the binomial coefficients $\binom{n}{k}$, $\binom{n}{k-1}$, and $\binom{n+1}{k}$ using Lucas' theorem. Observe that the first entry, the one corresponding to the 0-th power of two, in the binary expansion $\text{bin}(n)$ of n is zero since n is even.

- 1) Assume k is even. In this case, c is forced to be 0 since $k-1 \not\leq n$ (the first entry of $\text{bin}(k-1)$ is 1). This implies $a = b$. In other words, the only possible solutions are $(0, 0, 0)$ and $(\pm, \pm, 0)$. As $\text{bin}(n+1) = 1 \frown \text{bin}(n)^{\{0\}}$, we have $k \leq n$ if and only if $k \leq n+1$, thus each of these solutions is in fact consistent.
- 2) Assume k is odd. Since $k \not\leq n$, $b = 0$ (the first entry of $\text{bin}(k)$ is 1). This implies $a = c$. In other words, the only possible solutions are $(0, 0, 0)$ and $(\pm, 0, \pm)$. Again, each of these solutions is consistent due to $\text{bin}(n+1) = 1 \frown \text{bin}(n)^{\{0\}}$ and $\text{bin}(k) = 1 \frown \text{bin}(k-1)^{\{0\}}$, so that $k-1 \leq n$ if and only if $k \leq n+1$. $\square$

Proposition 4.3 *Let $n \in \mathbb{N}$ be odd and $f_0, f_1 \in \mathcal{B}_n$. Define $f \in \mathcal{B}_{n+1}$ to be the concatenation of f_0 with f_1 , i.e. $f = f_0 || f_1$. The following hold for $k \in [2, n-1]$ and $(a, b, c) \in \{0, +, -\}^3$.*

- 1) *If k is even, then only for either $(a, b, c) = (0, 0, 0)$, $(a, b, c) = (\pm, \pm, 0)$, or $(a, b, c) = (0, \pm, \mp)$, it holds*

$$f \in \mathcal{WAPB}_{k,n+1}^a \text{ if } f_0 \in \mathcal{WAPB}_{k,n}^b \text{ and } f_1 \in \mathcal{WAPB}_{k-1,n}^c.$$

Moreover, the solution $(a, b, c) = (\pm, 0, \pm)$ is also possible if $k \equiv 0 \pmod{2^{l_0}}$, where 2^{l_0} is the largest power of two that divides $n+1$.

- 2) *If k is odd, then only for either $(a, b, c) = (0, 0, 0)$ or $(a, b, c) = (0, \pm, \mp)$, it holds*

$$f \in \mathcal{WAPB}_{k,n+1}^a \text{ if } f_0 \in \mathcal{WAPB}_{k,n}^b \text{ and } f_1 \in \mathcal{WAPB}_{k-1,n}^c.$$

Proof. Suppose that $f_0 \in \mathcal{WAPB}_{k,n}^b$ and $f_1 \in \mathcal{WAPB}_{k-1,n}^c$. Similarly, as in the proof of Prop. 4.2, we get $|\text{supp}_k(f)| = \frac{\binom{n+1}{k} + a}{2}$, where $a := b + c$. In this case, the first entry in the binary expansion $\text{bin}(n)$ is one since n is odd. Let l_0 be the first zero entry in $\text{bin}(n)$. This value corresponds to the entry indexed by 2^{l_0} . Clearly,

$$\text{bin}(n+1) = \underbrace{(0, 0, \dots, 0, 1)}_{l_0 \text{ times}} \frown \text{bin}(n)^{[0, l_0]}. \quad (12)$$

- 1) Assume k is even. Suppose that $b = 0$. Then, $a = c$, which in turn implies that $(0, 0, 0)$ and $(\pm, 0, \pm)$ are possible solutions. The solution $(a, b, c) = (\pm, 0, \pm)$ is however impossible unless $k \equiv 0 \pmod{2^{l_0}}$. Indeed, let x be an index witnessing $k \not\leq n$. By definition of l_0 , $x > l_0$. This readily implies that x is also a witness of $k \not\leq n+1$ by (12), concluding that $(a, b, c) = (\pm, 0, \pm)$ cannot happen in this case. Now, W.L.O.G, suppose that $b = 1$ (the case $b = -1$ is symmetric), i.e., $k \leq n$. From $a = b + c$, we get that either $a = 1, c = 0$ or $a = 0, c = -1$. Let us see if these are consistent solutions. Let l'_0 be the first zero entry in $\text{bin}(k-1)$, as before,

$$\text{bin}(k) = \underbrace{(0, 0, \dots, 0, 1)}_{l'_0 \text{ times}} \frown \text{bin}(k-1)^{[0, l'_0]}. \quad (13)$$

It must be that $l_0 \neq l'_0$ by $k \leq n$. If $l_0 < l'_0$, then the entries at l_0 of $\text{bin}(n+1)$, $\text{bin}(n)$, $\text{bin}(k)$ and $\text{bin}(k-1)$ are, respectively, 1, 0, 0, and 1. This means that $k-1 \not\leq n$. Moreover, $k \leq n+1$ since $k \leq n$ and the vector $\text{bin}(n)$ is equal to $\text{bin}(n+1)$ for every index greater than l_0 . Thus, $a = 1$ and $c = 0$. If $l_0 > l'_0$, then the entries at l'_0 of $\text{bin}(n+1)$, $\text{bin}(n)$, $\text{bin}(k)$, and $\text{bin}(k-1)$ are, respectively, 0, 1, 1, and 0. This implies that $k \not\leq n+1$. Finally, $k-1 \leq n$ since $k \leq n$ and all entries in $\text{bin}(n)$ smaller than l_0 are 1. Consequently, $a = 0$ and $c = -1$.

- 2) Assume k is odd. In this case, $k \not\leq n+1$. This implies $a = 0$, so that $b = -c$. In other words, the only possible solutions can be $(0, 0, 0)$ and $(0, \pm, \mp)$. It is now clear that $k \leq n$ if and only if $k-1 \leq n$. This yields that these are the only consistent solutions. $\square$

Example 4.1 Let $f_0, f_1 \in \mathcal{B}_5$ be given as

$$f_0(x) = x_0x_1x_2x_3x_4 + x_0x_1x_2 + x_0x_2x_3 + x_0x_2x_4 + x_1x_3x_4 + x_0x_2 + x_0 + x_1 + x_4 + 1$$

and

$$f_1(x) = x_0x_1x_2x_3x_4 + x_0x_1x_2x_3 + x_0x_1x_2 + x_0x_2x_3 + x_0x_2x_4 + x_1x_3x_4 + x_1x_3 + x_0 + x_1 + x_4 + 1.$$

Both functions f_0, f_1 are WAPB, with the supp_k distribution given in Table 1.

Table 1: Weightwise support distribution of f_0 and f_1 .

k	$ E_{k,n} $	$ \text{supp}_k(f_0) $	$ \text{supp}_k(f_1) $
1	5	2	2
2	10	5	5
3	10	5	5
4	5	2	3

Then, by Proposition 4.3, the following holds for the concatenated function $\mathcal{B}_6 \ni f = f_0 || f_1$:

$$k = 2: \quad f \in \mathcal{WAPB}_{2,6}^- \quad \text{since} \quad f_0 \in \mathcal{WAPB}_{2,5}^0 \quad \text{and} \quad f_1 \in \mathcal{WAPB}_{1,5}^-,$$

$$k = 3: \quad f \in \mathcal{WAPB}_{3,6}^0 \quad \text{since} \quad f_0 \in \mathcal{WAPB}_{3,5}^0 \quad \text{and} \quad f_1 \in \mathcal{WAPB}_{2,5}^0,$$

$$k = 4: \quad f \in \mathcal{WAPB}_{4,6}^- \quad \text{since} \quad f_0 \in \mathcal{WAPB}_{4,5}^- \quad \text{and} \quad f_1 \in \mathcal{WAPB}_{3,5}^0.$$

So, the function f is 2-WAPB⁽⁻¹⁾, 3-WAPB⁽⁰⁾, and 4-WAPB⁽⁻¹⁾. Moreover, it is 1-WAPB⁽⁰⁾ and 5-WAPB⁽⁰⁾, and hence it is WAPB.

5 WPB functions from the GMM class

Definition 5.1 Let n, r, s be positive integers such that $n = r + s$, the general Maiorana-McFarland class, denoted by $\mathcal{GMM}_n$, contains all n -variable Boolean functions of the form

$$f(x, y) = x \cdot \varphi(y) + g(y), \quad x \in \mathbb{F}_2^r, \quad y \in \mathbb{F}_2^s, \quad (14)$$

where $\varphi: \mathbb{F}_2^s \rightarrow \mathbb{F}_2^r$, and g is a Boolean function in $\mathcal{B}_s$.

Note that the Walsh transform of a function $f \in \mathcal{GMM}_n$ at a point $(u, v) \in \mathbb{F}_2^r \times \mathbb{F}_2^s$ is given by

$$W_f(u, v) = \sum_{y \in \mathbb{F}_2^s} (-1)^{g(y) + v \cdot y} \sum_{x \in \mathbb{F}_2^r} (-1)^{x \cdot \varphi(y) + u \cdot x}.$$

In this context, the slices $E_{k,n}$ are clearly equal to $\bigcup_{i+j=k} E_{j,r} \times E_{i,s}$, where i and j are such that $i \leq s, j \leq r$. When restricting the domain to the slices $E_{k,n}$, we get the following equation, for each $k \in [1, n-1]$:

$$W_{f,k}(u, v) = \sum_{(x,y) \in E_{k,n}} (-1)^{x \cdot \varphi(y) + g(y) + (u,v) \cdot (x,y)} = \sum_{i+j=k} \sum_{y \in E_{i,s}} (-1)^{g(y) + v \cdot y} \sum_{x \in E_{j,r}} (-1)^{x \cdot \varphi(y) + u \cdot x}. \quad (15)$$

In the particular case of $(u, v) = (0_r, 0_s)$, we get

$$W_{f,k}(0_r, 0_s) = \sum_{i+j=k} \sum_{y \in E_{i,s}} (-1)^{g(y)} \sum_{x \in E_{j,r}} (-1)^{x \cdot \varphi(y)}. \quad (16)$$

Equivalently,

$$|\text{supp}_k(f)| = 2^{n-1} - \frac{1}{2} \sum_{i+j=k} \sum_{y \in E_{i,s}} (-1)^{g(y)} \sum_{x \in E_{j,r}} (-1)^{x \cdot \varphi(y)}.$$

For any function $\varphi : \mathbb{F}_2^s \rightarrow \mathbb{F}_2^r$ and for $l \in [0, s]$, define the vector

$$\nu^{l,\varphi} = (\nu_y)_{y \in E_{l,s}} \in [0, s]^{\binom{s}{l}}$$

by $\nu^{l,\varphi} := (wt(\varphi(y)))_{y \in E_{l,s}}$. From Lemma 2.1 and (16), we get

$$W_{f,k}(0_r, 0_s) = \sum_{i+j=k} \sum_{y \in E_{j,s}} (-1)^{g(y)} P_i(\nu_y, r). \quad (17)$$

It is evident now that the key to finding W(A)PB functions in the GMM class lies in the evaluation of Krawtchouk polynomials at weights of images of the function φ , and how well these behave with the function $g(y)$ in order to annihilate the sum

$$\sum_{i+j=k} \sum_{y \in E_{j,s}} (-1)^{g(y)} P_i(\nu_y, r).$$

The goal is then to find all or some possible solutions for $\nu^{l,\varphi}$ for a given g using known symmetries of Krawtchouk polynomials. To fix ideas, let us compute the possible values for $r = s = 4$, so that $n = 8$. Note that there are $5^{\sum_{i=0}^4 \binom{4}{i}} = 5^{16} \approx 2^{37}$ choices for $\nu^{l,\varphi}$ and 2^{16} choices for g , thus we must impose some restrictions on g and find effective reductions that allow us compute a subset of suitable values.

In general, one possible reduction is achieved by letting g be a simple function, denoted by h , such as the indicator of $\mathbb{F}_2^{n-r} \setminus \{0\}$, i.e. let $h : \mathbb{F}_2^{n-r} \rightarrow \mathbb{F}_2$ be defined as

$$h(y) = \begin{cases} 1, & \forall y \in \mathbb{F}_2^{n-r} \setminus \{0\}, \\ 0, & \text{if } y = 0. \end{cases} \quad (18)$$

For $n = 4$, from (17), the following equations are obtained. For $k = 1$, we have

$$W_{f,1}(0_4, 0_4) = \sum_{i+j=1} \sum_{y \in E_{j,4}} (-1)^{g(y)} P_i(\nu_y, 4) = (-1)^{h(0_4)} P_1(\nu_{0_4}, 4) + \sum_{y \in E_{1,4}} (-1)^{h(y)} P_0(\nu_y, 4).$$

Using the definition of h and $P_0(x) = 1$ for each x , this can be turned into

$$P_1(\nu_{0_4}, 4) - \sum_{y \in E_{1,4}} P_0(\nu_y, 4) = P_1(\nu_{0_4}, 4) - 4.$$

Therefore, our first constraint is $P_1(\nu_{0_4}, 4) = 4$, so that $\nu_{0_4} = 0$, equivalently, $\varphi(0_4) = 0_4$. Similarly, for $k = 2$, the Walsh coefficient $W_{f,2}(0_4, 0_4)$ equals

$$P_2(\nu_{0_4}, 4) + \sum_{y \in E_{1,4}} (-1)^{h(y)} P_1(\nu_y, 4) + \sum_{y \in E_{2,4}} (-1)^{h(y)} P_0(\nu_y, 4) = - \sum_{y \in E_{1,4}} P_1(\nu_y, 4).$$

This yields the condition $\sum_{y \in E_{1,4}} P_1(\nu_y, 4) = 0$. In general, notice that

$$P_l(\nu_{0_4}, 4) + \sum_{y \in E_{l,4}} (-1)^{h(y)} P_0(\nu_y, 4) = 0.$$

For $k \in [3, 7]$, the equations are

$$\begin{aligned} \sum_{y \in E_{2,4}} P_1(\nu_y, 4) + \sum_{y \in E_{1,4}} P_2(\nu_y, 4) &= 0, \\ \sum_{y \in E_{3,4}} P_1(\nu_y, 4) + \sum_{y \in E_{2,4}} P_2(\nu_y, 4) + \sum_{y \in E_{1,4}} P_3(\nu_y, 4) &= 0 \\ P_1(\nu_{1_4}, 4) + \sum_{y \in E_{3,4}} P_2(\nu_y, 4) + \sum_{y \in E_{2,4}} P_3(\nu_y, 4) + \sum_{y \in E_{1,4}} P_4(\nu_y, 4) &= 0 \\ P_2(\nu_{1_4}, 4) + \sum_{y \in E_{3,4}} P_3(\nu_y, 4) + \sum_{y \in E_{2,4}} P_4(\nu_y, 4) &= 0 \\ P_3(\nu_{1_4}, 4) + \sum_{y \in E_{3,4}} P_4(\nu_y, 4) &= 0 \end{aligned}$$

Moreover, Since $f(0_4, 0_4) = 0$, it must be that $\varphi(1_4)$ is even (so that $f(1_4, 1_4) = 1$). These equations suggest a bottom-up, a top-down or a hybrid dynamic programming approach to find the possible assignments of values for $\nu^{l,\varphi}$, see Algorithm 1 in Appendix A.

Example 5.1 For $n = 8$ and $r = 4$, we give φ in its lookup table representation as

$$[\varphi(0, 0, 0, 0), \varphi(1, 0, 0, 0), \dots, \varphi(1, 1, 1, 1)].$$

For $\varphi_1 = [0, 15, 12, 15, 4, 14, 6, 5, 8, 9, 10, 8, 2, 1, 0, 3]$, we define $f_1(x, y) = x \cdot \varphi_1(y) + h(y)$, and, for $\varphi_2 = [0, 13, 10, 11, 12, 5, 6, 7, 4, 9, 2, 12, 0, 3, 1, 10]$, we define $f_2(x, y) = x \cdot \varphi_2(y) + h(y)$. The functions φ_1 and φ_2 were obtained using Algorithm 1. Consequently, the functions f_1 and f_2 are WPB, each having nonlinearity 112, which represents the best results achieved by Algorithm 1. From Table 2, we notice that f_1 and f_2 compare well with the other known WPB functions in 8-variables in terms of NL_k .

Unfortunately, Algorithm 1 is infeasible for $r = s = 8$ (i.e. $n = 16$). To obtain WPB functions in the GMM class, a slightly different approach can be put forth to reevaluate the

¹An exhaustive search of all functions defined over $E_{2,8}$ (resp. on $E_{6,8}$) was performed to verify that the best 2-weight nonlinearity (resp. 6-weight nonlinearity) is 9.

Table 2: The k -weight nonlinearity of known 8-variable WPB functions.

Functions	[16]	[22]	[26]	[31]	[23]	[31]	[21]	[21]	[10]	f_1	f_2	Upper Bound
NL ₂ ¹	2	2	2	2	6	6	4	4	9	6	6	11
NL ₃	12	14	12	12	8	17	12	8	22	18	16	24
NL ₄	19	19	19	19	26	23	20	18	27	23	25	30
NL ₅	12	14	12	12	8	17	14	8	22	16	18	24
NL ₆	2	2	2	6	6	6	6	6	9	7	5	11

Table 3: Krawtchouk matrix of order 8.

	$P_i(0)$	$P_i(1)$	$P_i(2)$	$P_i(3)$	$P_i(4)$	$P_i(5)$	$P_i(6)$	$P_i(7)$	$P_i(8)$
$i = 0$	1	1	1	1	1	1	1	1	1
$i = 1$	8	6	4	2	0	-2	-4	-6	-8
$i = 2$	28	14	4	-2	-4	-2	4	14	28
$i = 3$	56	14	-4	-6	0	6	4	-14	-56
$i = 4$	70	0	-10	0	6	0	-10	0	70
$i = 5$	56	-14	-4	6	0	-6	4	14	-56
$i = 6$	28	-14	4	2	-4	2	4	-14	28
$i = 7$	8	-6	4	-2	0	2	-4	6	-8
$i = 8$	1	-1	1	-1	1	-1	1	-1	1

combined role that g and φ play in (14). Let $r = s = 8$ and consider the Krawtchouk matrix of order 8 given in Table 3. Define

$$A_i = \sum_{y \in E_{i,8}} (-1)^{g(y)}.$$

Upon examining vlaues in Table 3, it is natural to exploit the symmetry of rows. We do so by searching for values that make $A_i = 0$ for i odd. For a vector in $\mathbb{F}_2^8 \times \mathbb{F}_2^8$ of weight $k \in \{1, \dots, 15\}$, compute all corresponding equations derived from (17). For instance, for $k = 1$,

$$(-1)^{g(0_8)} P_1(\nu_{0_8}, 8) + \sum_{y \in E_{1,8}} (-1)^{g(y)} = (-1)^{g(0_8)} P_1(\nu_{0_8}, 8) + A_1 = 0,$$

and, for $k = 15$,

$$(-1)^{g(1_8)} P_7(\nu_{1_8}, 8) + \sum_{y \in E_{7,8}} (-1)^{g(y)} P_8(\nu_y) = 0.$$

Since we want to get $A_1 = 0$ and $A_7 = 0$, one can impose the constraints: $g(0_8) = g(1_8) = 0$, $\nu_{0_8} = \nu_{1_8} = (4)$, and, let all entries in $\nu^{7,\varphi}$ have the same parity. Furthermore, to simplify other expressions, let $\nu^{i,\varphi}$ be constant for each odd index i . For $k = 2$, we then have

$$(-1)^{g(0_8)} P_2(\nu_{0_8}, 8) + \sum_{y \in E_{1,8}} (-1)^{g(y)} P_1(\nu_y) + A_2 = 0.$$

Replacing the corresponding values yields $A_2 = 4$. The unknown vectors $\nu^{2,\varphi}$, $\nu^{4,\varphi}$ and $\nu^{6,\varphi}$ are restricted by A_2 , A_4 and A_6 , respectively. Moreover, these vectors must be chosen suitably since their values affect some of the other equations. For instance, since $A_2 = 4$, the entries of $\nu^{2,\varphi}$ have to be chosen such that h is balanced on the corresponding 24 vectors of weight 2 and maps the other 4 vectors of weight 2 to 0. At the same time, we must assure that $\nu^{2,\varphi}$ satisfies as many equations as possible for $k = 3, 4, \dots, 10$.

A partial solution is to let $\nu^{2,\varphi}$ have 25 entries with 2, 2 entries with 4 and 1 entry with 6. Similarly, we can obtain $A_4 = -6$ and $A_6 = 8$ by selecting $\nu^{4,\varphi}$ and $\nu^{6,\varphi}$ appropriately. To avoid displaying lengthy vectors, let us use the compact notation for the value distribution of $\nu^{i,\varphi}$,

$$\text{dist}(\nu^{i,\varphi}) = a_0^{\pm e_1} a_1^{\pm e_2} \dots a_t^{\pm e_t},$$

where $a_j \in [0, 8]$ and $e_j \in \binom{[8]}{i}$ indicate that the value a_j appears e_j times in $\nu^{i,\varphi}$ and h must map to either 0 or 1 on the corresponding vector depending on whether there is a plus or a minus sign. Our discussion above results in

$$\text{dist}(\nu^{2,\varphi}) = 2^{-12} 2^{13} 4^2 6^1, \quad \text{dist}(\nu^{4,\varphi}) = 2^{-1} 3^{-1} 4^{-34} 4^{34} 5^{-1} 6^{-1}$$

and $\text{dist}(\nu^{6,\varphi}) = 2^1 3^2 4^2 5^2 6^{-10} 6^{11}$. This selection of values yields a function that is close to being WPB—the function is balanced on all slices, but two. Namely, the function is almost balanced on the slices $k = 10$ and $k = 14$, therefore we only have to correct these slices by swapping the values of two vectors in both of them. This leads to a 16-variable weightwise perfect balanced function in the GMM class, whose ANF is given in Appendix B and its slice nonlinearities are displayed in Table 4.

Table 4: The k -slice nonlinearity of known 16-variable WPB functions.

Function	[16]	[22]	[26]	[31]	[31]	[21]	[21]	f	Upper bound
NL ₂	4	4	4	4	12	40	38	34	54
NL ₃	56	112	56	56	104	204	219	211	268
NL ₄	350	686	350	350	590	765	758	794	888
NL ₅	1312	1806	1288	1288	1765	1814	1887	2023	2150
NL ₆	3176	3436	3108	3108	3487	3484	3511	3785	3959
NL ₇	4782	4994	4774	4774	5154	5138	5041	5473	5666
NL ₈	5443	5603	5539	5539	5827	5875	5699	6160	6378
NL ₉	4782	4994	4902	4902	5154	5158	5105	5508	5666
NL ₁₀	3176	3436	3236	3228	3491	3514	3601	3822	3959
NL ₁₁	1312	1806	1672	1664	1765	1819	1879	2051	2150
NL ₁₂	350	686	654	638	590	743	702	808	888
NL ₁₃	56	112	152	152	104	182	189	206	268
NL ₁₄	4	4	28	12	12	35	36	30	54

Remark 5.1 Note that the function $f = x \cdot \varphi(y) + g(y)$ in Table 4 supersedes all known WPB functions in terms of their weightwise nonlinearities. Furthermore, its nonlinearity is 32508, which is one of the best nonlinearities exhibited by a WPB function in 16 variables.

6 Equivalence of WAPB functions

As shown in [10], adding a symmetric function, i.e., a linear combination of indicators of slices, to a WAPB function does not affect the weightwise balanced property and it moreover preserves the slice nonlinearities of the function. Thus, one may consider such functions as equivalent. More generally, consider a bijective linear transformation L . We say that L is *slice-invariant* if either it fixes slices set-wise or swaps them symmetrically, i.e., for each $k \in [1, n-1]$, $L(E_{k,n}) = E_{k,n}$ or $L(E_{k,n}) = E_{n-k,n}$. Observe that there are $\frac{(2n!)}{n!}$ slice-invariant linear permutations.

Definition 6.1 *Two functions $f, f' : \mathbb{F}_2^n \rightarrow \mathbb{F}_2$ are said to be slice-equivalent if there exist a slice-invariant linear permutation $L : \mathbb{F}_2^n \rightarrow \mathbb{F}_2^n$ and a symmetric function $\sigma : \mathbb{F}_2^n \rightarrow \mathbb{F}_2$ such that $f'(x) = f(L(x)) + \sigma(x)$ for each x . Denote this equivalence relation by $\sim$.*

Proposition 6.1 *Suppose that $f', f \in \mathcal{B}_n$ are functions with $f \sim f'$, so that $f'(x) = f(L(x)) + \sigma(x)$ for a slice-invariant linear permutation L and a symmetric function $\sigma = \sum_{j \in [0, n]} \lambda_j \chi_{j,n}$. Then, for each $k \in [n-1]$ and $\omega \in \mathbb{F}_2^n$, either*

$$W_{f',k}(\omega) = (-1)^{\lambda_k} W_{k,f}((L^{-1})^*(\omega)) \text{ or } W_{f',k}(\omega) = (-1)^{\lambda_k} W_{n-k,f}((L^{-1})^*(\omega)),$$

where $(L^{-1})^*$ denotes the adjoint operator of L^{-1} . Therefore,

$$NL_k(f') = NL_k(f) \text{ or } NL_k(f') = NL_{n-k}(f)$$

for each k , depending on the action of L . Moreover, if f is WAPB, then so is f' .

Proof. Let f, f', L and σ as in the statement. For a given $k \in [n-1]$ and $\omega \in \mathbb{F}_2^n$, $W_{f',k}(\omega) = \sum_{x \in E_{k,n}} (-1)^{f'(x) + \omega \cdot x} = \sum_{x \in E_{k,n}} (-1)^{f(L(x)) + \sigma(x) + \omega \cdot x}$. Then,

$$W_{f',k}(\omega) = \sum_{y \in L(E_{k,n})} (-1)^{f(y) + \sigma(L^{-1}(y)) + \omega \cdot L^{-1}(y)} = (-1)^{\lambda_k} \sum_{y \in L(E_{k,n})} (-1)^{f(y) + \omega \cdot L^{-1}(y)}.$$

The latter is equal to either

$$(-1)^{\lambda_k} \sum_{y \in E_{k,n}} (-1)^{f(y) + (L^{-1})^*(\omega) \cdot y} = (-1)^{\lambda_k} W_{k,f}((L^{-1})^*(\omega))$$

or $(-1)^{\lambda_k} \sum_{y \in E_{n-k,n}} (-1)^{f(y) + (L^{-1})^*(\omega) \cdot y} = (-1)^{\lambda_k} W_{n-k,f}((L^{-1})^*(\omega))$. It follows easily that if f is WAPB then so is f' and that the slice nonlinearities are preserved. $\square$

Remark 6.1 *It was essentially proved in [10] that every WAPB function f is slice-equivalent to a special WAPB function, where the witnesses are the identity function on $\mathbb{F}_2^n$ and the symmetric function σ that “corrects” the signs of the Walsh values, namely, if*

$$S = \{k \in [0, \lfloor \frac{n-1}{2} \rfloor] : W_{f,k}(0_n) = -1\} \cup \{k \in [\lceil \frac{n+1}{2} \rceil, n] : W_{f,k}(0_n) = 1\},$$

then $f + \sum_{i \in S} \chi_{i,n}$ is SWAPB.

6.1 A simple construction method from restrictions

In what follows, we present a rather straightforward approach, based on restrictions of functions, to build WAPB functions. Given $n - 1$ restricted-domain functions $f_1, \dots, f_{n-1}$, $f_i : E_{i,n} \rightarrow \mathbb{F}_2$, one can define a function $F : \mathbb{F}_2^n \rightarrow \mathbb{F}_2$ by letting $F \upharpoonright_{E_{i,n}} = f_i$ and $F(0_n), F(1_n) \in \mathbb{F}_2$. Consider the extensions F_i of each function f_i to the whole space $\mathbb{F}_2^n$ given by

$$F_i(x) = \begin{cases} f_i(x), & x \in E_{i,n}, \\ 0, & \text{otherwise.} \end{cases} \quad (19)$$

The nonlinearity of the overall function can be bounded below by the slice nonlinearities of the restriction functions. Namely, we have the following.

Proposition 6.2 *Let $f_1, \dots, f_{n-1}$ be non-constant functions $f_i : E_{i,n} \rightarrow \mathbb{F}_2$. If the function $F : \mathbb{F}_2^n \rightarrow \mathbb{F}_2$ is given by $F \upharpoonright_{E_{i,n}} = f_i$, then*

$$W_F(v) = \sum_{i=1}^{n-1} W_{i,F_i}(v) + (-1)^{F(1_n)+v \cdot 1_n} + (-1)^{F(0_n)},$$

where the extension functions F_i are defined in (19). Moreover, $NL(F) \geq \sum_{i=1}^{n-1} NL_i(F_i)^2$. In particular, if $F_i \sim F_{n-i}$ for every i , then

$$NL(F) \geq 2 \sum_{i=1}^{T-1} NL_i(F_i) + (\delta + 1)NL_T(F_T),$$

where $T := \lfloor \frac{n}{2} \rfloor$ and $\delta \in \{0, 1\}$ is the value of n modulo 2.

Proof. The first part is trivial, so we only show the second part. Taking absolute value of the Walsh coefficients given in the first part, we get

$$|W_F(v)| = \left| \sum_{i=1}^{n-1} W_{i,F_i}(v) + (-1)^{F(1_n)+v \cdot 1_n} + (-1)^{F(0_n)} \right| \leq \sum_{i=1}^{n-1} |W_{i,F_i}(v)| + 2$$

by the triangle inequality. Considering the maximum over $v \in \mathbb{F}_2^n$ yields

$$\max_{v \in \mathbb{F}_2^n} |W_F(v)| \leq \max_{v \in \mathbb{F}_2^n} \left(\sum_{i=1}^{n-1} |W_{i,F_i}(v)| + 2 \right) = \sum_{i=1}^{n-1} \max_{v \in \mathbb{F}_2^n} |W_{i,F_i}(v)| + 2.$$

Hence, $NL(F)$ is at least

$$2^{n-1} - \frac{1}{2} \sum_{i=1}^{n-1} \max_{v \in \mathbb{F}_2^n} |W_{i,F_i}(v)| - 1 = \frac{1}{2} \sum_{i=1}^{n-1} \left(\binom{n}{i} - \max_{v \in \mathbb{F}_2^n} |W_{i,F_i}(v)| \right) = \sum_{i=1}^{n-1} NL_i(F_i).$$

²In [11], this number was called the global weightwise nonlinearity, denoted by $\text{GWNL}(f)$. Proposition 6.2 shows a slightly more general result than Proposition 1 in [11].

Now, suppose that the extensions F_i satisfy $F_i \sim F_{n-i}$ for every $i \in [n-1]$. For each $1 \leq i \leq \lfloor \frac{n+1}{2} \rfloor$, let L_i be a slice-invariant linear permutation on $\mathbb{F}_2^n$ and $\sigma_i \in \mathcal{B}_n$ be a symmetric function such that $F_{n-i}(x) = F_i(L_i(x)) + \sigma_i(x)$. Since $\text{supp}(F_i) \subset E_{i,n}$ and $\text{supp}(F_{n-i}) \subset E_{n-i,n}$, we have $L_i(E_{i,n}) = E_{n-i}$ (and vice versa). Then, Proposition 6.1 readily yields $NL(F) \geq 2 \sum_{i=1}^{T-1} NL_i(F_i) + (\delta+1)NL_T(F_T)$. $\square$

Theorem 6.1 *Let $f_1, \dots, f_{n-1}$ be non-constant restricted-domain functions $f_i : E_{i,n} \rightarrow \mathbb{F}_2$. Consider the function $F : \mathbb{F}_2^n \rightarrow \mathbb{F}_2$ given by $F \upharpoonright_{E_{i,n}} = f_i$. If F_i is i -WAPB for every i , then F is a WAPB function.*

Proof. For each $i \in [n-1]$, since F_i is i -WAPB,

$$W_{i,F}(0_n) = \sum_{x \in E_{i,n}} (-1)^{F(x)} = \sum_{x \in E_{i,n}} (-1)^{f_i(x)} = \frac{\binom{n}{i} + \varepsilon}{2},$$

where $\varepsilon = 0$ when $\binom{n}{i} \equiv 0 \pmod{2}$ and $\varepsilon \in \{-1, 1\}$ otherwise. Therefore, F is WAPB. $\square$

Corollary 1 *Let $f_1, \dots, f_{n-1}$ be non-constant restricted-domain functions $f_i : E_{i,n} \rightarrow \mathbb{F}_2$. Consider the function $F : \mathbb{F}_2^n \rightarrow \mathbb{F}_2$ given by $F \upharpoonright_{E_{i,n}} = f_i$. Let*

$$S := \{(i, n-i) : i \in [\lfloor \frac{n-1}{2} \rfloor]\}$$

and let τ be a function whose domain is S and $\tau(x) \in x$ for $x \in S$. If the extensions F_i in (19) are such that $F_i \sim F_{n-i}$ and $F_{\tau(x)}$ is $\tau(x)$ -WAPB for every $x \in S$, then F is a WAPB function.

Proof. As in the proof of Proposition 6.2, we know that a slice-invariant linear permutation for the equivalence between F_i and F_{n-i} has to symmetrically swap E_i and E_{n-i} . Using Proposition 6.1, we get that $F_{n-\tau(x)}$ is $(n-\tau(x))$ -WAPB as $F_{\tau(x)}$ is $\tau(x)$ -WAPB for every $x \in S$. Theorem 6.1 now yields the result. $\square$

Another easy way to build functions satisfying the conditions in Corollary 1 is obtained when τ is the function selecting the left and right coordinate of pairs in S by comparing the weightwise nonlinearities on the symmetric slices. Namely, given a function $f : \mathbb{F}_2^n \rightarrow \mathbb{F}_2$, its *mirror function* $\hat{f} : \mathbb{F}_2^n \rightarrow \mathbb{F}_2$ is the function

$$\hat{f}(x) = \begin{cases} f(\bar{x}), & NL_k(f) < NL_{n-k}(f), \text{ for } k \in [2, n-2], \\ f(x), & \text{otherwise.} \end{cases}$$

Remark 6.2 *These two approaches allow construction of $W(A)PB$ functions with high weightwise nonlinearities by combining the optimal weightwise nonlinearities of several $W(A)PB$ functions. Note that the mirroring approach has been implicitly used before (see [10]).*

Example 6.1 Let $G \in \mathcal{B}_{16}$ be given by

$$G \upharpoonright_{E_{i,16}} = \begin{cases} c_1 \upharpoonright_{E_{i,16}}, & \text{for } i = 2, 3 \\ f \upharpoonright_{E_{i,16}}, & \text{for } 0 \leq i \leq 16, i \neq 2, 3, 14 \\ c_2 \upharpoonright_{E_{14,16}}, & \end{cases}$$

where $c_1, c_2 \in \mathcal{B}_{16}$ are functions described in [21] in Table 6, with parameters (u, t, s) equal to $(0, 0, 128), (1, 1, 111)$, respectively, and $f \in \mathcal{B}_{16}$ is the function presented in Table 4. Function G is obtained by patching together the slices with the highest nonlinearities of functions c_1, c_2 and f , whereas function $\hat{G}$ is obtained by mirroring the slices with higher nonlinearities of G . The k -slice and overall nonlinearities of G and $\hat{G}$ and the restricted-domain functions are given in Table 5.

Table 5: Nonlinearities of functions G and $\hat{G}$, where G is obtained using Theorem 6.1 for the slices of c_1, c_2 and f from Table 4, and $\hat{G}$ is its mirror function.

	NL ₂	NL ₃	NL ₄	NL ₅	NL ₆	NL ₇	NL ₈	NL ₉	NL ₁₀	NL ₁₁	NL ₁₂	NL ₁₃	NL ₁₄	NL
c_1 [21]	40	204	765	1814	3484	5138	5875	5158	3514	1819	743	182	35	30306
c_2 [21]	38	219	758	1887	3511	5041	5699	5105	3601	1879	702	189	36	30332
f	34	211	794	2023	3785	5473	6160	5508	3822	2051	808	206	30	32508
G	40	219	794	2023	3785	5473	6160	5508	3822	2051	808	206	36	32378
$\hat{G}$	40	219	808	2051	3822	5508	6160	5508	3822	2051	808	219	40	32154

7 Conclusion

In this paper, we introduced two general constructions of weightwise (almost) perfectly balanced (W(A)PB) functions that achieve high overall and weightwise nonlinearities. The first construction, based on the concatenation of Boolean functions, refines and generalizes the ideas from [10] by providing a combinatorial characterization of when the concatenation of WPB functions results in a W(A)PB function. We further analyzed the general case of concatenations of WAPB functions and derived sufficient conditions under which they preserve the WAPB property, obtaining functions with high overall and slice nonlinearities.

The second construction, based on the general Maiorana–McFarland class, allowed us to explicitly determine the slice nonlinearities of the resulting W(A)PB functions and to study their attainable values for dimensions $n \leq 16$. We demonstrated that functions within this class can reach or even surpass the best-known nonlinearities reported in the literature to date, thereby confirming the strength of our approach.

Moreover, we introduced the concept of slice equivalence, which accounts for the effect of linear transformations that either fix or symmetrically swap slices. This concept of equivalence allowed us to propose a simple yet effective approach for generating new W(A)PB functions, namely, patching together restricted-domain functions that already exhibit high weightwise nonlinearity. In addition, mirroring the symmetric slices with higher weightwise nonlinearities allowed us to obtain a function in 16 variables which has the best individual

k -slice nonlinearities and its overall nonlinearity supersedes majority of the known 16-variable WPB functions.

Overall, our results provide a coherent framework for constructing and analyzing W(A)PB functions with high nonlinearities. Future work may focus on tightening the nonlinearity bounds for the proposed constructions and determining whether they can reach theoretical optima. Moreover, investigating alternative selections of functions g and φ within the GMM framework could result in new W(A)PB functions exhibiting even higher nonlinearities and additional desirable cryptographic properties.

Lastly, a detailed study of the number of distinct W(A)PB functions obtained up to slice equivalence would provide insights into how many of these constructions yield genuinely new functions and how much redundancy exists between the proposed classes and known families of W(A)PB functions.

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9 Statements and Declarations

The authors have no competing interests.

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A Algorithms

Algorithm 1 Find all tuples $\nu^{i,\varphi}$, $0 \leq i \leq s$ for which $f(x, y) = x \cdot \varphi(y) + h(y)$ is WPB.

Input r, s , sets of equations $E_1, \dots, E_{r+s-1}$

Output Set of possible values $\nu^{i,\varphi}$

```

1: define  $n \leftarrow r + s$ 
2: procedure UNORDEREDTUPLES( $l, E, array, left, index, current, tuples$ )
3:   define  $N \leftarrow \binom{s}{l} + s$ 
4:   if  $left = 0$  then
5:     if  $current$  satisfies equations in  $E$  then
6:       append  $current$  to  $tuples$ 
7:     end if
8:   end if
9:   for  $i := index$  do  $N - 1$ 
10:    append  $array[i]$  to  $current$ 
11:    UNORDEREDTUPLES( $l, E, array, left - 1, i + 1, current, tuples$ )
12:    remove last element in  $current$ 
13:  end for
14:  return  $tuples$ 
15: end procedure

16: procedure FIND( )
17:    $filteredTuples \leftarrow$  UNORDEREDTUPLES( $1, E_1, [], n, 0, [], \emptyset$ )
18:   for  $i := 2$  do  $n - 1$ 
19:     for  $j \in filteredTuples$  do
20:        $res \leftarrow \emptyset$ 
21:       append UNORDEREDTUPLES( $i, E_i, [], n, 0, [], res$ ) to  $j$ 
22:     end for
23:   end for
24:   return  $filteredTuples$ 
25: end procedure

```

B ANF of 16-variable WPB function

Table 6: Each hexadecimal string $d_1d_2 \cdots d_\delta$ represents a monomial of degree δ in the ANF, where each digit d_i denotes the variable x_{d_i} . In other words, the string encodes the product $x_{d_1}x_{d_2} \cdots x_{d_\delta}$. For example, the hexadecimal string **189ABC** corresponds to the term of degree 6 given as $x_1x_8x_9x_{10}x_{11}x_{12}$.

Deg.	Terms
15	0123456789ABCDEF, 0123456789ABCEFF, 012345689ABCDEF, 012345789ABCDEF
14	0123456789ABCD, 0123456789ABCE, 012345689ABCDE, 012345689ABCDF, 012345689ABCEFF, 012345689ABDEF, 012345689ACDEF, 012345789ABCDE, 012345789ABCDF, 012345789ABCEFF, 012345789ABDEF, 012345789ACDEF
13	012345689ABCD, 012345689ABCE, 012345689ABCF, 012345689ABDE, 012345689ABDF, 012345689ABEF, 012345689ACDE, 012345689ACDF, 012345689ACEF, 012345689ADEF, 012345789ABCD, 012345789ABCE, 012345789ABCF, 012345789ABDE, 012345789ABDF, 012345789ABEF, 012345789ACDE, 012345789ACDF, 012345789ACEF, 012345789ADEF
12	012345689ABC, 012345689ABD, 012345689ABE, 012345689ABF, 012345689ACD, 012345689ACE, 012345689ACF, 012345689ADE, 012345689ADF, 012345689AEF, 012345789ABC, 012345789ABD, 012345789ABE, 012345789ABF, 012345789ACD, 012345789ACE, 012345789ACF, 012345789ADE, 012345789ADF, 012345789AEF
11	012345689AB, 012345689AC, 012345689AD, 012345689AE, 012345689AF, 012345789AB, 012345789AC, 012345789AD, 012345789AE, 012345789AF
10	012345689A, 012345789A
9	089ABCDEF, 389ABCDEF, 489ABCDEF, 589ABCDEF, 689ABCDEF, 789ABCDEF
8	189ABCDE, 189ABCDF, 189ABDEF, 189ACDEF, 189BCDEF, 18ABCDEF, 289ABDEF, 289ACDEF, 28ABCDEF, 29ABCDEF, 389ABCEF, 389ACDEF, 489ABCEF, 48ABCDEF, 589ABCDE, 589ABCDF, 589ABCEF, 589ABDEF, 589BCDEF, 59ABCDEF, 689ABCDE, 689ABCDF, 689ABCEF, 689BCDEF, 68ABCDEF, 69ABCDEF, 789ABCDE, 789ABCDF, 789ABDEF, 789ACDEF, 789BCDEF, 78ABCDEF, 89ABCDEF
7	189ABCD, 189ABCF, 189ABDE, 189ACDE, 189BCDE, 18ABCDE, 19ABCDF, 19ABCEF, 289ABDE, 289ABEF, 289ACDE, 289ACEF, 289BCDF, 289BCEF, 28ABCDE, 29ABCDEF, 29ABCDF, 29ABCEF, 29ABDEF, 29ACDEF, 389ABCE, 389ABCF, 389ABDF, 389ABEF, 389ACDE, 389ADEF, 389BCDF, 389BCEF, 389BDEF, 389CDEF, 38ABDEF, 38ACDEF, 39ABCDF, 39ABCEF, 39ABDEF, 39ACDEF, 489ABCE, 489ABCF, 489ADEF, 489BDEF, 489CDEF, 48ABCDE, 48ABCDF, 48ABCEF, 48BCDEF, 49BCDEF, 4ABCDEF, 589ABCD, 589ABCE, 589ABCF, 589ABDE, 589ABDF, 589ACEF, 589BCDE, 589BCDF, 589BCEF, 58BCDEF, 59ABCDE, 59ABCDF, 59ABCEF, 59ABDEF, 59ACDEF, 59BCDEF, 5ABCDEF, 689ABCD, 689ABCE, 689ABEF, 689ACEF, 689ADEF, 689BCDE, 689BCDF, 689BCEF, 689BDEF, 689CDEF, 68ABCDE, 68BCDEF, 69ABCDE, 69ABCDF, 69ABCEF, 69ABDEF, 69ACDEF, 69BCDEF, 6ABCDEF, 789ABCD, 789ABDE, 789ABEF, 789ACDE, 789ACDF, 789ADEF, 789BCDE, 789BCDF, 789BCEF, 789BDEF, 789CDEF, 78ABCDE, 78ABCDF, 78ABCEF, 78ABDEF, 78ACDEF, 78BCDEF, 79ABCDF, 79ABCEF, 79BCDEF, 7ABCDEF, 89ABCDE, 89BCDEF, 8ABCDEF, 9ABCDEF

6	189ABC, 19ABCD, 19ABCE, 19ABCF, 289ABE, 289ABF, 289ACE, 289ACF, 289AEF, 289BCD, 289BCE, 289BCF, 29ABCD, 29ABCE, 29ABCF, 29ABDE, 29ABDF, 29ABEF, 29ACDE, 29ACDF, 29ACEF, 29ADEF, 389ABC, 389ABD, 389ABE, 389ABF, 389ADE, 389ADF, 389AEF, 389BCD, 389BCE, 389BCF, 389BDE, 389BDF, 389BEF, 389CDE, 389CDF, 389CEF, 389DEF, 38ABDE, 38ABDF, 38ABEF, 38ACDE, 38ACDF, 38ACEF, 38ADEF, 39ABCD, 39ABCE, 39ABCF, 39ABDE, 39ABDF, 39ABEF, 39ACDE, 39ACDF, 39ACEF, 39ADEF, 489ABC, 489ADE, 489ADF, 489AEF, 489BDE, 489BDF, 489BEF, 489CDE, 489CDF, 489CEF, 489DEF, 48ABCD, 48ABCE, 48ABCF, 48BCDE, 48BCDF, 48BCEF, 48BDEF, 48CDEF, 49BCDE, 49BCDF, 49BCEF, 49BDEF, 49CDEF, 4ABCDE, 4ABCDF, 4ABCEF, 4ABDEF, 4ACDEF, 4BCDEF, 589ABC, 589ABD, 589ACE, 589ACF, 589AEF, 589BCD, 589BCE, 589BCF, 58BCDE, 58BCDF, 58BCEF, 58BDEF, 58CDEF, 59ABCD, 59ABCE, 59ABCF, 59ABDE, 59ABDF, 59ABEF, 59ACDE, 59ACDF, 59ACEF, 59ADEF, 59BCDE, 59BCDF, 59BCEF, 59BDEF, 59CDEF, 5ABCDE, 5ABCDF, 5ABCEF, 5ABDEF, 5ACDEF, 5BCDEF, 689ABE, 689ABF, 689ACE, 689ACF, 689ADE, 689ADF, 689BCD, 689BCE, 689BCF, 689BDE, 689BDF, 689BEF, 689CDE, 689CDF, 689CEF, 689DEF, 68BCDE, 68BCDF, 68BCEF, 68BDEF, 68CDEF, 69ABCD, 69ABCE, 69ABCF, 69ABDE, 69ABDF, 69ABEF, 69ACDE, 69ACDF, 69ACEF, 69ADEF, 69BCDE, 69BCDF, 69BCEF, 69BDEF, 69CDEF, 6ABCDE, 6ABCDF, 6ABCEF, 6ABDEF, 6ACDEF, 6BCDEF, 789ABE, 789ABF, 789ACD, 789ADE, 789ADF, 789BCD, 789BCE, 789BCF, 789BDE, 789BDF, 789BEF, 789CDE, 789CDF, 789CEF, 789DEF, 78ABCD, 78ABCE, 78ABCF, 78ABDE, 78ABDF, 78ABEF, 78ACDE, 78ACDF, 78ACEF, 78ADEF, 78BCDE, 78BCDF, 78BCEF, 78BDEF, 78CDEF, 79ABCD, 79ABCE, 79ABCF, 79BCDE, 79BCDF, 79BCEF, 79BDEF, 79CDEF, 7ABCDE, 7ABCDF, 7ABCEF, 7ABDEF, 7ACDEF, 7BCDEF, 89ABCD, 89ABCF, 89ABDF, 89ABEF, 89ACDF, 89ACEF, 89BCDF, 89BDEF, 8ABCDE, 8ACDEF, 8BCDEF, 9ABDEF, 9ACDEF, ABCDEF
5	19ABC, 289AB, 289AC, 289AE, 289AF, 289BC, 29ABC, 29ABD, 29ABE, 29ABF, 29ACD, 29ACE, 29ACF, 29ADE, 29ADF, 29AEF, 389AB, 389AD, 389AE, 389AF, 389BC, 389BD, 389BE, 389BF, 389CD, 389CE, 389CF, 389DE, 389DF, 389EF, 38ABD, 38ABE, 38ABF, 38ACD, 38ACE, 38ACF, 38ADE, 38ADF, 38AEF, 39ABC, 39ABD, 39ABE, 39ABF, 39ACD, 39ACE, 39ACF, 39ADE, 39ADF, 39AEF, 489AD, 489AE, 489AF, 489BD, 489BE, 489BF, 489CD, 489CE, 489CF, 489DE, 489DF, 489EF, 48ABC, 48BCD, 48BCE, 48BCF, 48BDE, 48BDF, 48BEF, 48CDE, 48CDF, 48CEF, 48DEF, 49BCD, 49BCE, 49BCF, 49BDE, 49BDF, 49BEF, 49CDE, 49CDF, 49CEF, 49DEF, 4ABCD, 4ABCE, 4ABCF, 4ABDE, 4ABDF, 4ABEF, 4ACDE, 4ACDF, 4ACEF, 4ADEF, 4BCDE, 4BCDF, 4BCEF, 4BDEF, 4CDEF, 589AC, 589AE, 589AF, 589BC, 58BCD, 58BCE, 58BCF, 58BDE, 58BDF, 58BEF, 58CDE, 58CDF, 58CEF, 58DEF, 59ABC, 59ABD, 59ABE, 59ABF, 59ACD, 59ACE, 59ACF, 59ADE, 59ADF, 59AEF, 59BCD, 59BCE, 59BCF, 59BDE, 59BDF, 59BEF, 59CDE, 59CDF, 59CEF, 59DEF, 5ABCD, 5ABCE, 5ABCF, 5ABDE, 5ABDF, 5ABEF, 5ACDE, 5ACDF, 5ACEF, 5ADEF, 5BCDE, 5BCDF, 5BCEF, 5BDEF, 5CDEF, 689AB, 689AC, 689AD, 689BC, 689BD, 689BE, 689BF, 689CD, 689CE, 689CF, 689DE, 689DF, 689EF, 68BCD, 68BCE, 68BCF, 68BDE, 68BDF, 68BEF, 68CDE, 68CDF, 68CEF, 68DEF, 69ABC, 69ABD, 69ABE, 69ABF, 69ACD, 69ACE, 69ACF, 69ADE, 69ADF, 69AEF, 69BCD, 69BCE, 69BCF, 69BDE, 69BDF, 69BEF, 69CDE, 69CDF, 69CEF, 69DEF, 6ABCD, 6ABCE, 6ABCF, 6ABDE, 6ABDF, 6ABEF, 6ACDE, 6ACDF, 6ACEF, 6ADEF, 6BCDE, 6BCDF, 6BCEF, 6BDEF, 6CDEF, 789AB, 789AD, 789BC, 789BD, 789BE, 789BF, 789CD, 789CE, 789CF, 789DE, 789DF, 789EF, 78ABC, 78ABD, 78ABE, 78ABF, 78ACD, 78ACE, 78ACF, 78ADE, 78ADF, 78AEF, 78BCD, 78BCE, 78BCF, 78BDE, 78BDF, 78BEF, 78CDE, 78CDF, 78CEF, 78DEF, 79ABC, 79BCD, 79BCE, 79BCF, 79BDE, 79BDF, 79BEF, 79CDE, 79CDF, 79CEF, 79DEF, 7ABCD, 7ABCE, 7ABCF, 7ABDE, 7ABDF, 7ABEF, 7ACDE, 7ACDF, 7ACEF, 7ADEF, 7BCDE, 7BCDF, 7BCEF, 7BDEF, 7CDEF, 89ABE, 89ACD, 89ADE, 89ADF, 89BCD, 89BCF, 89BDE, 89DEF, 8ABCE, 8ABDF, 8ACDE, 8BCDE, 8BCEF, 8BDEF, 9ABCD, 9ABCE, 9ABCF, 9ABDE, 9ACDE, 9ADEF, 9BCEF, 9BDEF, 9CDEF, ABCDE, ABCDF, ABCEF, ABDEF, BCDEF

4	289A, 29AB, 29AC, 29AD, 29AE, 29AF, 389A, 389B, 389C, 389D, 389E, 389F, 38AB, 38AC, 38AD, 38AE, 38AF, 39AB, 39AC, 39AD, 39AE, 39AF, 489A, 489B, 489C, 489D, 489E, 489F, 48BC, 48BD, 48BE, 48BF, 48CD, 48CE, 48CF, 48DE, 48DF, 48EF, 49BC, 49BD, 49BE, 49BF, 49CD, 49CE, 49CF, 49DE, 49DF, 49EF, 4ABC, 4ABD, 4ABE, 4ABF, 4ACD, 4ACE, 4ACF, 4ADE, 4ADF, 4AEF, 4BCD, 4BCE, 4BCF, 4BDE, 4BDF, 4BEF, 4CDE, 4CDF, 4CEF, 4DEF, 589A, 58BC, 58BD, 58BE, 58BF, 58CD, 58CE, 58CF, 58DE, 58DF, 58EF, 59AB, 59AC, 59AD, 59AE, 59AF, 59BC, 59BD, 59BE, 59BF, 59CD, 59CE, 59CF, 59DE, 59DF, 59EF, 5ABC, 5ABD, 5ABE, 5ABF, 5ACD, 5ACE, 5ACF, 5ADE, 5ADF, 5AEF, 5BCD, 5BCE, 5BCF, 5BDE, 5BDF, 5BEF, 5CDE, 5CDF, 5CEF, 5DEF, 689B, 689C, 689D, 689E, 689F, 68BC, 68BD, 68BE, 68BF, 68CD, 68CE, 68CF, 68DE, 68DF, 68EF, 69AB, 69AC, 69AD, 69AE, 69AF, 69BC, 69BD, 69BE, 69BF, 69CD, 69CE, 69CF, 69DE, 69DF, 69EF, 6ABC, 6ABD, 6ABE, 6ABF, 6ACD, 6ACE, 6ACF, 6ADE, 6ADF, 6AEF, 6BCD, 6BCE, 6BCF, 6BDE, 6BDF, 6BEF, 6CDE, 6CDF, 6CEF, 6DEF, 789B, 789C, 789D, 789E, 789F, 78AB, 78AC, 78AD, 78AE, 78AF, 78BC, 78BD, 78BE, 78BF, 78CD, 78CE, 78CF, 78DE, 78DF, 78EF, 79BC, 79BD, 79BE, 79BF, 79CD, 79CE, 79CF, 79DE, 79DF, 79EF, 7ABC, 7ABD, 7ABE, 7ABF, 7ACD, 7ACE, 7ACF, 7ADE, 7ADF, 7AEF, 7BCD, 7BCE, 7BCF, 7BDE, 7BDF, 7BEF, 7CDE, 7CDF, 7CEF, 7DEF, 89AB, 89AC, 89BE, 89BF, 89CD, 89CE, 89CF, 89DE, 89DF, 8ACF, 8BDE, 8BEF, 9ABC, 9ABF, 9ACD, 9ACE, 9ADE, 9AEF, 9BCF, 9BDF, 9BEF, 9CDE, 9CDF, ABCD, ABCE, ACDF, ACEF, BCDF
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2	0F, 1E, 2D, 3C, 48, 49, 4A, 4C, 4D, 4E, 4F, 58, 59, 5B, 5C, 5D, 5E, 5F, 68, 6A, 6B, 6C, 6D, 6E, 6F, 79, 7A, 7B, 7C, 7D, 7E, 7F, 8A, 9A, 9B, 9D, 9E, 9F, AB, AC, AD, AE, AF, BE, CE, DE
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